

COMPUTATIONAL DISPLAYS

Computational Imaging – Spring 2024

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Slide credits

Many of these slides are based on the following content:

- G. Wetzstein, D. Lanman, D. Gutierrez, M. Hirsch. **Computational Displays**. ACM SIGGRAPH 2012 Course.
- F. Banterle, A. Artusi, T. Aydin, P. Didyk, E. Eisemann, D. Gutierrez, R. Mantiuk, K. Myszkowski. **Multidimensional Image Retargeting**. ACM SIGGRAPH Asia 2011 Course.
- B. Masia et al. **A Survey on Computational Displays**. Computers and Graphics 2013.

Overview

Basis:

- ✓ Digital photography (Lecture #1)
- ✓ Mathematical foundations (Lecture #2)

Transient imaging (Lecture #9)

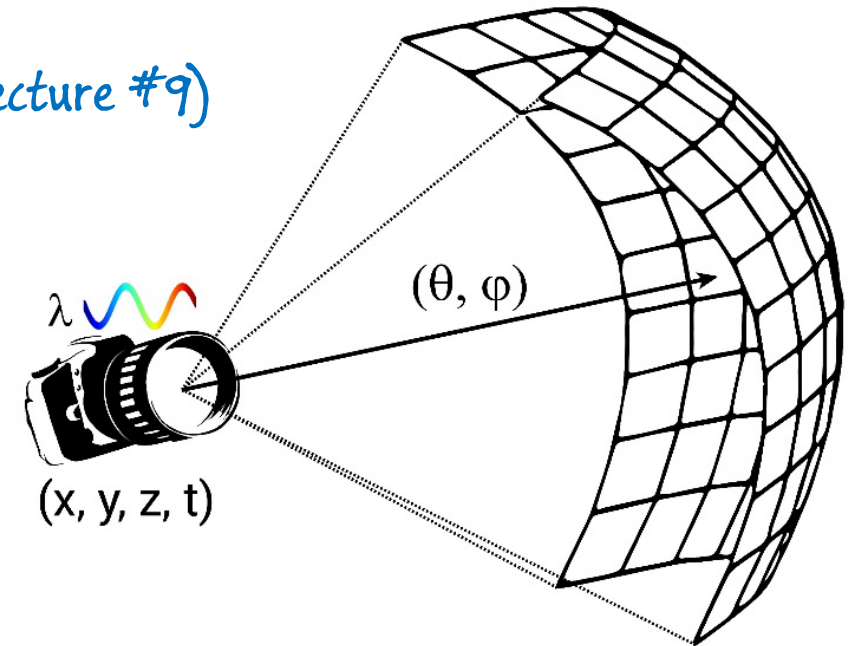
Hyperspectral imaging (Lecture #8)

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

✓ Light field imaging (Lecture #4)

✓ High dynamic range imaging (Lecture #5)

➡ + Computational displays (Lecture #6)



Plus common tools and techniques:

- ✓ Coded photography (Lecture #3)
- ✓ Compressive sensing (Lecture #3)
- Computational illumination (Lecture #7)

This is an imaging course... Why displays?

- One deals with acquisition of light, the other one with emission of light
- Both heavily related to optics, physics of light, computational illumination, image processing, ... → **Shared underlying principles**
- In many cases they are an integral part of the pipeline (contribute to visualization)

Computational displays

- Displays attempt to provide an **as-faithful-as-possible** reproduction of the real world (or of a virtual one)
- Today, they are still limited and can **not yet** provide images **indistinguishable from real world ones**.
- Doing so would require increasing:
 - Dynamic range
 - Color gamut
 - Spatial resolution
 - Temporal resolution
 - Angular resolution (stereo and light field displays)

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The plenoptic function:

$$P = P(\theta, \phi, \lambda, t, x, y, z)$$

High Dynamic Range (HDR) displays

HDR displays

- They seek to provide a large dynamic range.
- Dynamic range: Ratio between the maximum and minimum of the tonal values present in a scene/image/display...

Reminder:

The problem of dynamic range

- Our devices do not match the real world
- 10:1 photographic print (higher for glossy paper)
- 20:1 artist's paints
- 200:1 slide film
- 500:1 negative film
- 1,000:1 LCD display
- 2,000:1 digital SLR (at 12 bits)
- 100,000:1 real world

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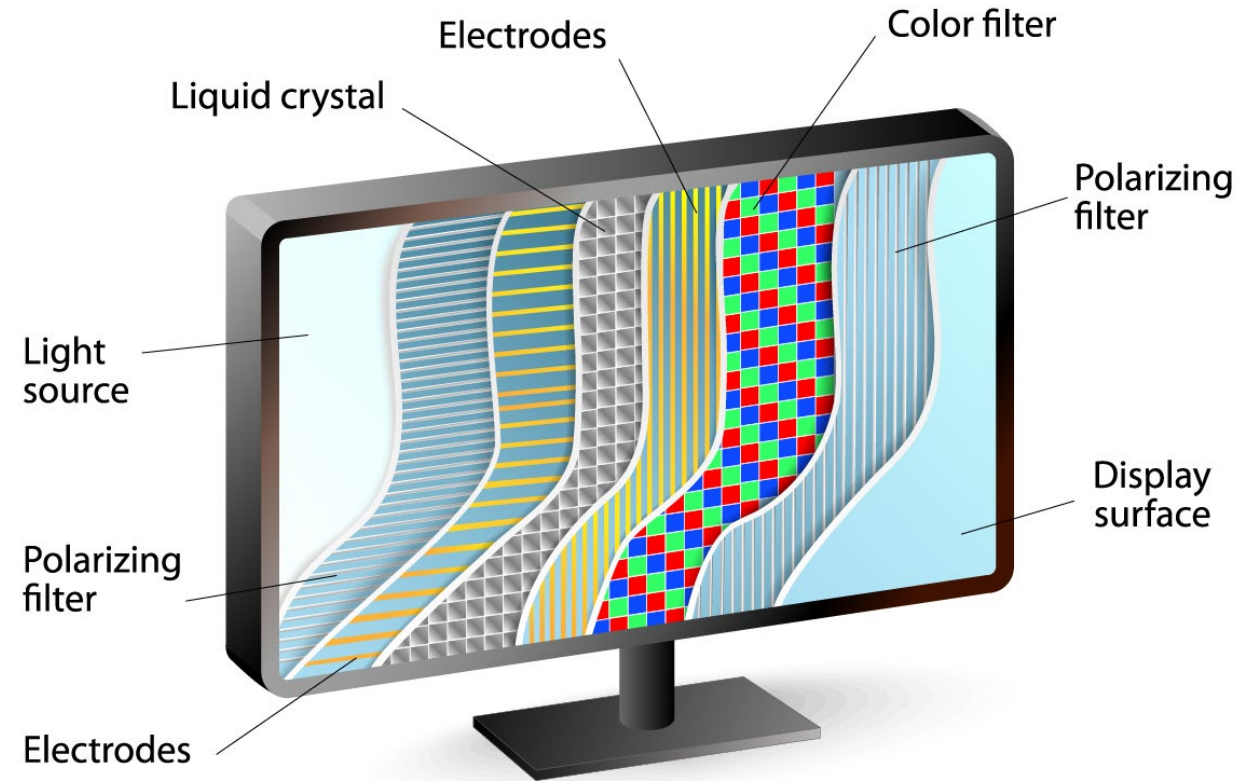
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HDR displays

- Hardware-based solutions
 - Dual modulation & backlight local dimming
- Software-based solutions
 - Apparent contrast enhancement
 - Binocular tone mapping

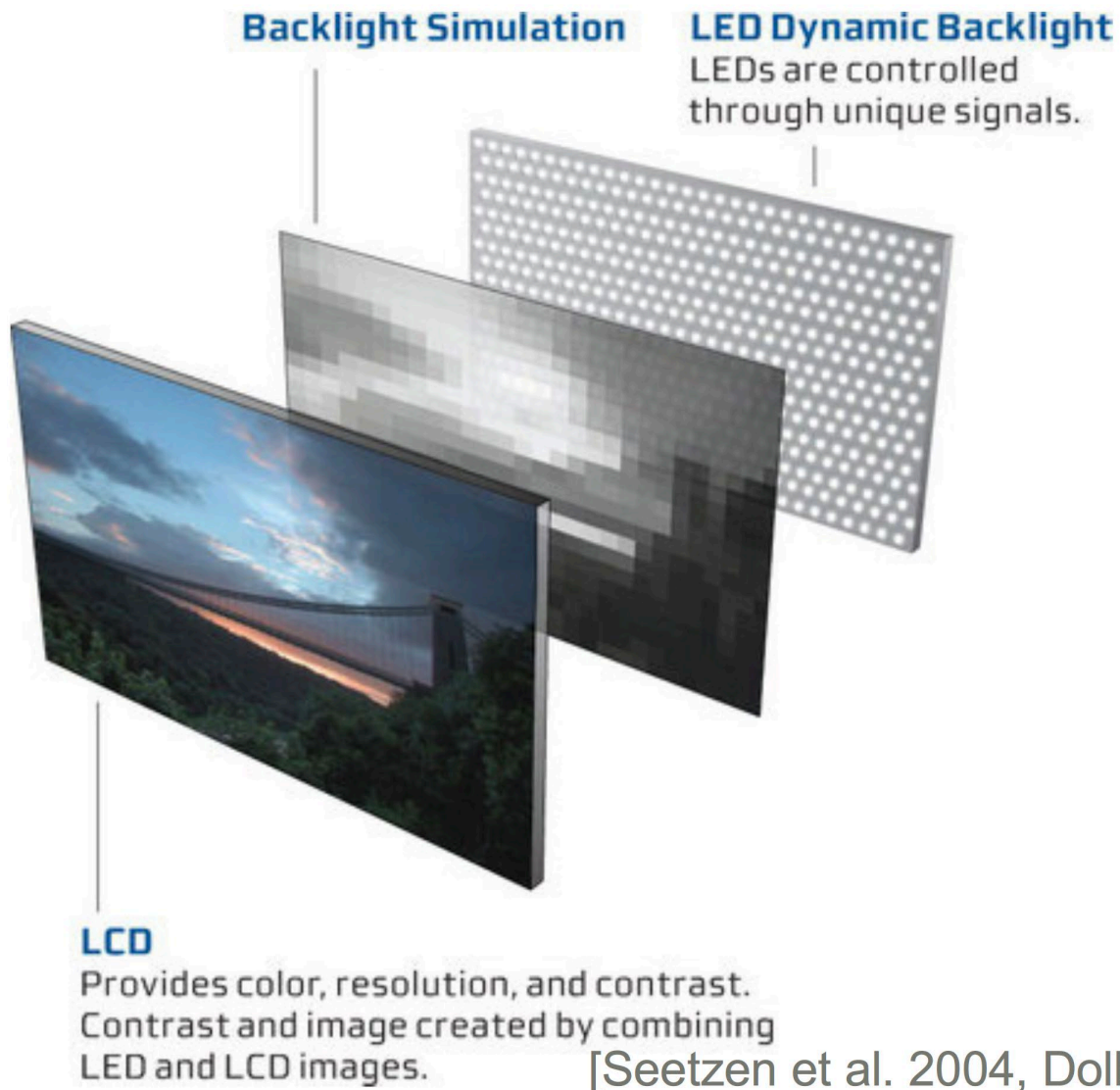
HDR displays – How do LCD displays work?

- Liquid crystals rotate the polarization of light depending on the voltage applied.
- Two linear polarizing filters, with liquid crystal cells in between, allow varying amounts of light to pass through depending on voltage.
- Often: A backlight acts as the light source, that is then modulated.
 - => Complicated to create blacks
 - => Reduced contrast



Source: Flexenable

HDR displays – Dual modulation



In practice, dual modulation is often:

LED backlight

+

LCD panel

Pre-computation (on device, real-time):

Target HDR image =

low-resolution, high contrast pattern (LEDs)

+

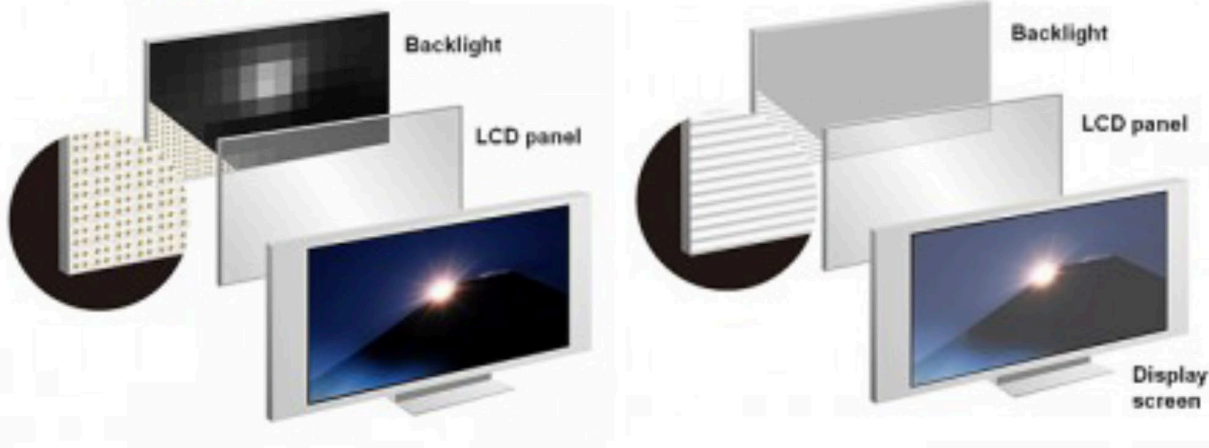
high-resolution image that adds sharp
image details and colors (LCD)

HDR displays – Backlight local dimming

Dual modulation (through backlight dimming) was also adopted to increase contrast in HDR content.

TRILUMINOS

RGB Dynamic LED



Display screen
with local
dimming
backlighting



Display screen
with CCFL
backlighting



Local dimming, Sony



Micro-dimming, Samsung

HDR displays – Backlight local dimming

Samsung, 85" 4K display with local dimming



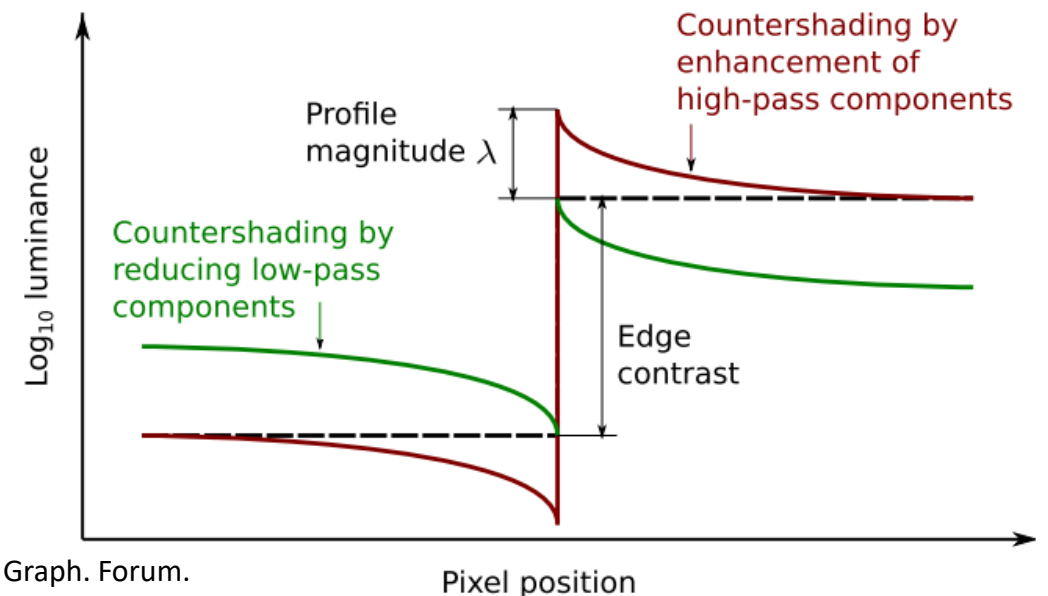
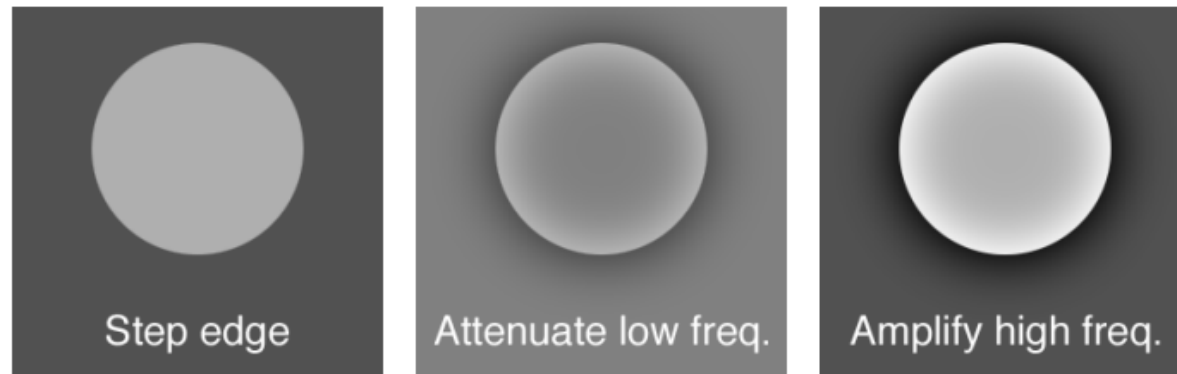
A wealth of techniques exist on how to compute the backlight dimming to offer the best trade-off between quality and energy consumption.

HDR displays

- Hardware-based solutions
 - Dual modulation & backlight local dimming
- Software-based solutions
 - Apparent contrast enhancement
 - Binocular tone mapping

HDR displays – Apparent contrast enhancement

- **Local contrast enhancement**: Image processing technique to enhance contrast.
- **Countershading**: Local edge contrast increased by adding/subtracting gradients to either side of the edge.
 - Amplifying the high-frequency components
 - Adding a high-pass image to the original image (often in image enhancement)
 - Reducing the low-frequency components
 - More often used in tone mapping



[Trentacoste et al. 2012] Unsharp masking, countershading and halos: enhancements or artifacts? Comp. Graph. Forum.

HDR displays – Binocular tone mapping

- From an HDR, two LDRs are generated, and presented for binocular fusion (**stereo viewing needed**).
- When the two images are presented simultaneously to both eyes (one to each eye), the fused image presents more detail than any of the two individual, low dynamic range depictions.



(a) Monocular LDR



(b) Binocular LDR pair

[Yang et al. 2012] Binocular tone mapping. ACM Trans. on Graph. & [Zhang et al. 2018] Binocular Tone Mapping with Improved Overall Contrast and Local Details. Comp. Graph. Forum.

HDR displays – Binocular tone mapping

* Assumption that large visual difference indicates more visual content.

- How is the pair obtained?

Given an LDR rendition, seek (optimization) an LDR image tone-mapped from the HDR input that:

- Has the largest visual difference with the existing one*, while
- preserving binocular fusibility for the LDR image pair.

(variations of this have been presented)



[Yang et al. 2012] Binocular tone mapping. ACM Trans. on Graph. & [Zhang et al. 2018] Binocular Tone Mapping with Improved Overall Contrast and Local Details. Comp. Graph. Forum.

Wide color gamut displays

Increasing color gamut

- **Color gamut:** Subset of colors that can be accurately represented in a given circumstance (e.g., reproducible by a display or representable by a color space).

- Color = lightness + chromaticity

Can be specified by (x,y) values.

Lightness:

Perception of a color's reflectance (not linear).

Reflectance:

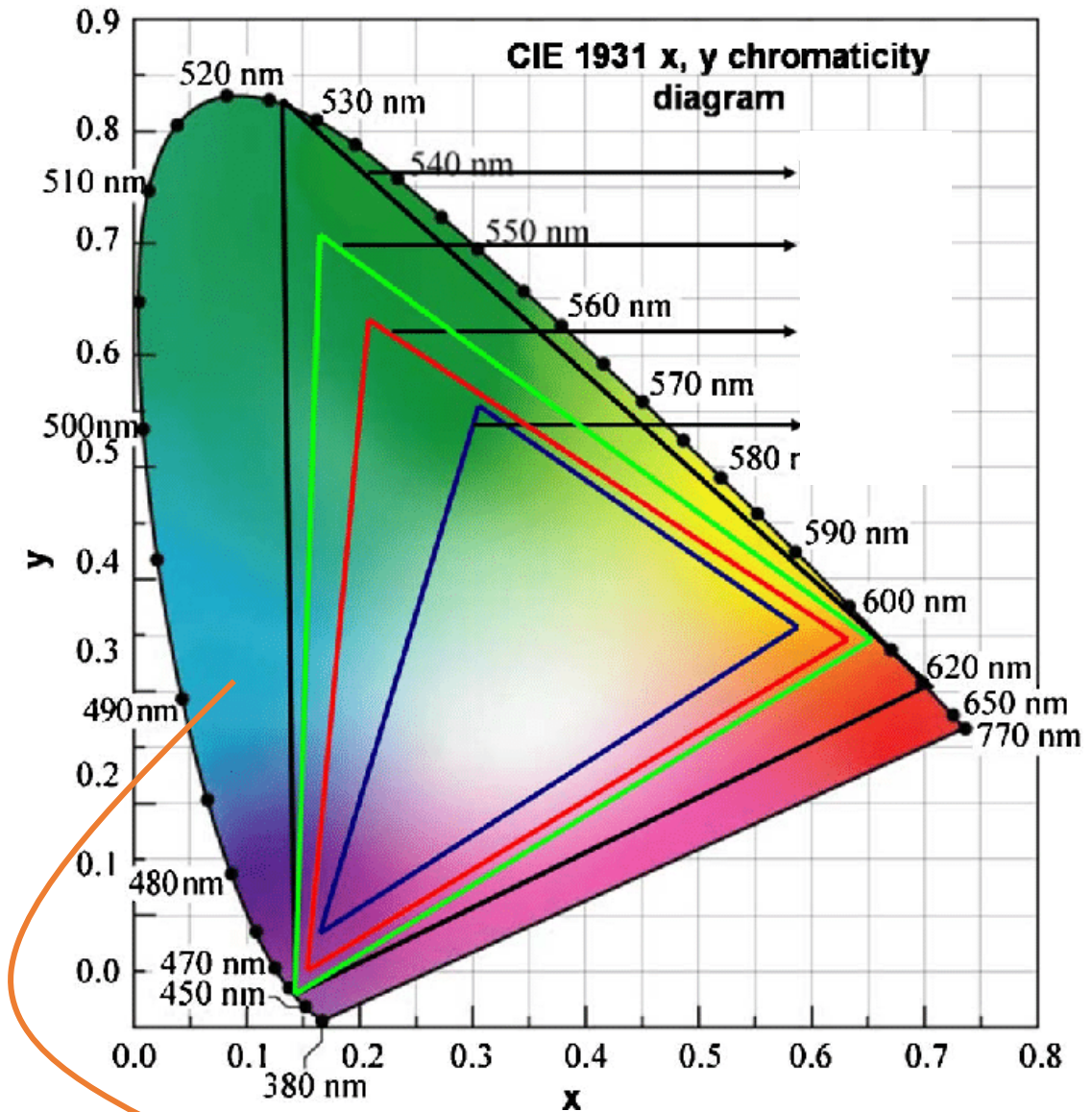
How much light is reflected from a surface.

Brightness:

Subjective impression/perception of luminance (not linear).

Luminance:

Luminous intensity per unit area (cd/m^2).

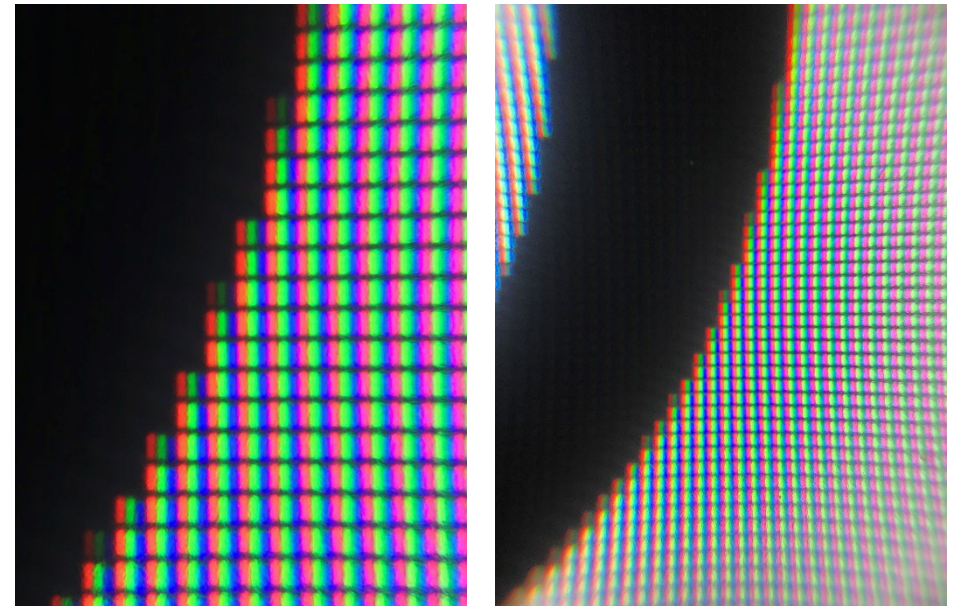
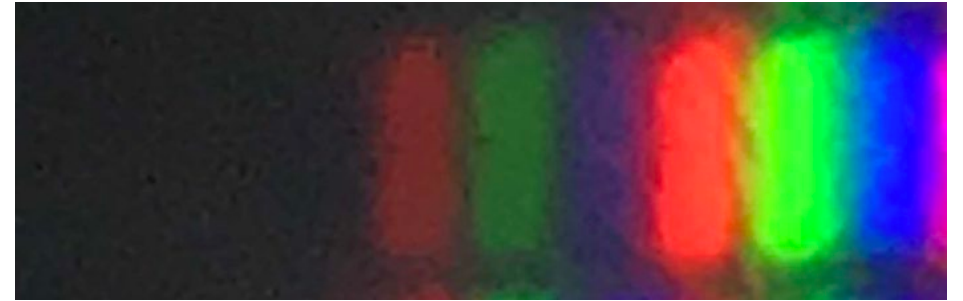


Human gamut:

All chromaticities visible to the average HVS.

Increasing color gamut

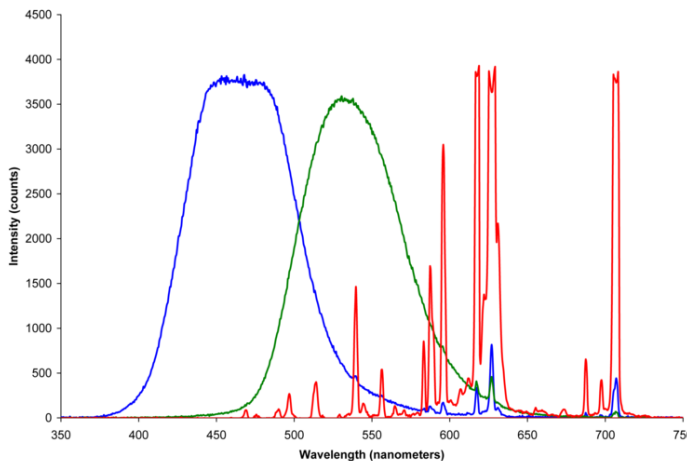
- **Color primaries:** Set of colored lights (emitting elements) that are mixed in varying amounts to produce a gamut.
- Multiple emitting elements co-stimulating the same area of the retina: **additive** behavior (i.e., predicted via summing SPDs).



Source: Makers Portal

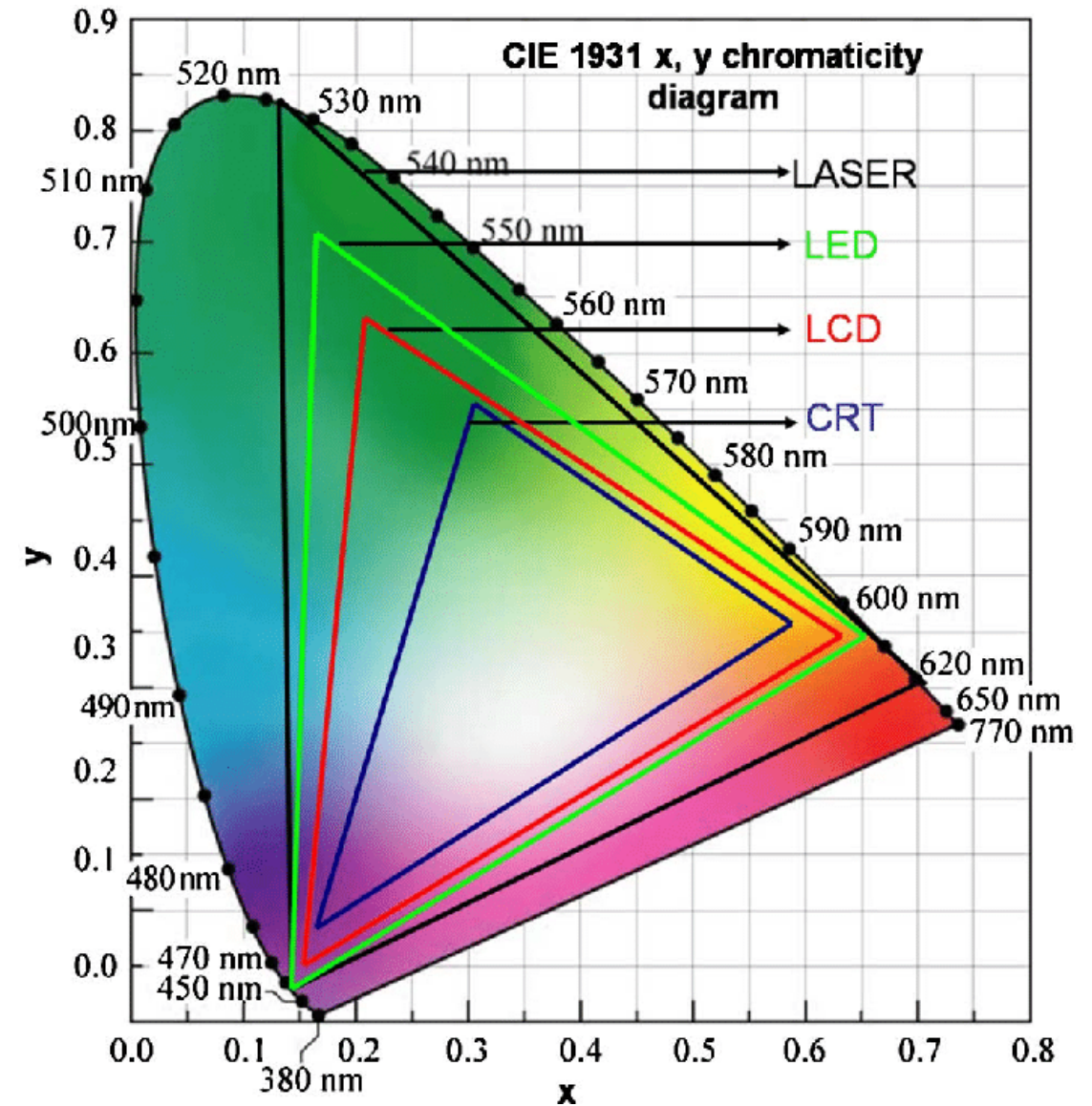
Increasing color gamut

- **Color primaries:** Set of colored lights (emitting elements) that are mixed in varying amounts to produce a gamut.
- Multiple emitting elements co-stimulating the same area of the retina: **additive** behavior (i.e., predicted via summing SPDs).
- Broad spectral distribution of the emitting elements limits the achievable gamut.



Emission spectra of the three phosphors that define the additive primary colors of a CRT display.

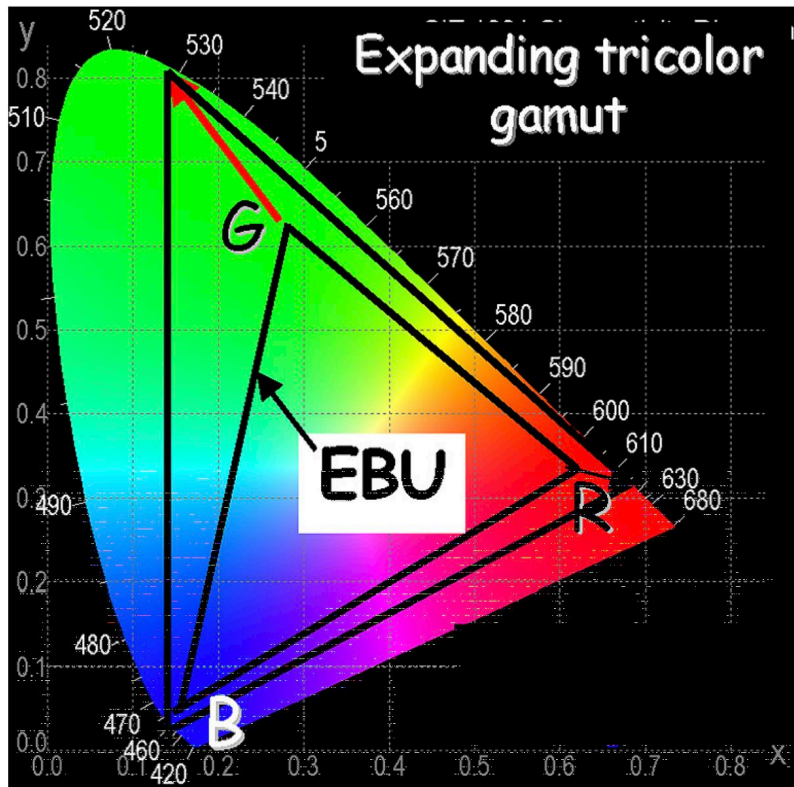
Source: Wikipedia.



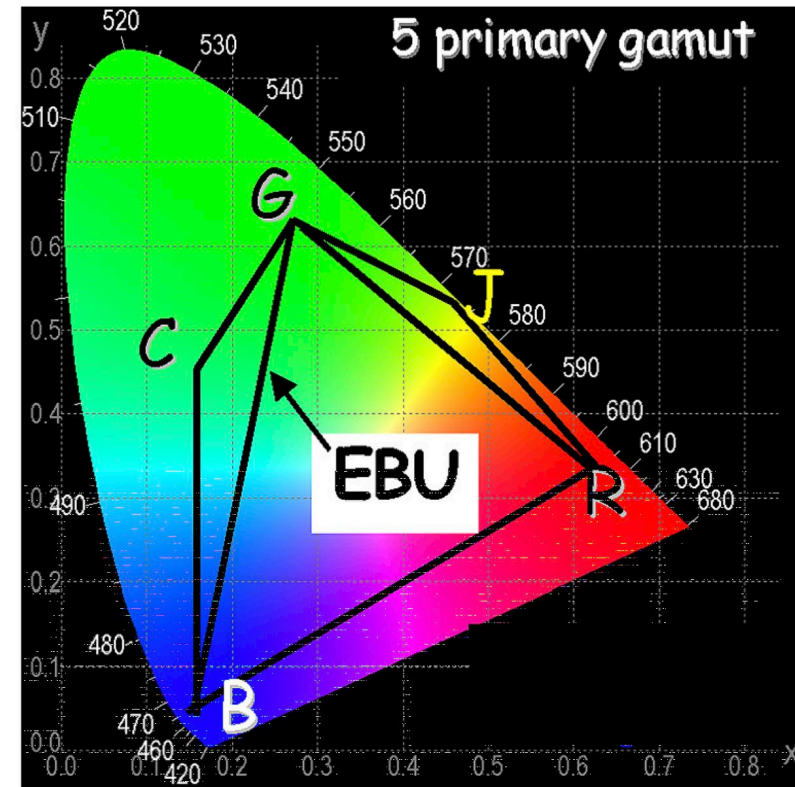
[Chellappan et al. 2010] Laser-based displays: A review. Applied Optics.

Increasing color gamut

Two basic ways of improving color gamut:



Use of more saturated primaries
(More saturation = Narrower
spectral width of the emitted light)



Use of more primaries
(Use of 4, 5 and 6 primaries has
been proposed. In this example,
yellow (J) and cyan (C))

Spatial superresolution displays

Increasing spatial resolution

Trivial answer: Pack more pixels!

Besides this, two types of techniques:

Increasing spatial resolution

Trivial answer: Pack more pixels!

Besides this, two types of techniques:

- Optical superposition:
 - Projection principle.
 - Low-resolution images from multiple devices are optically superimposed on a projection screen.
 - Superimposed images are shifted by some amount with respect to each other => one super-resolved pixel receives contributions from multiple devices.

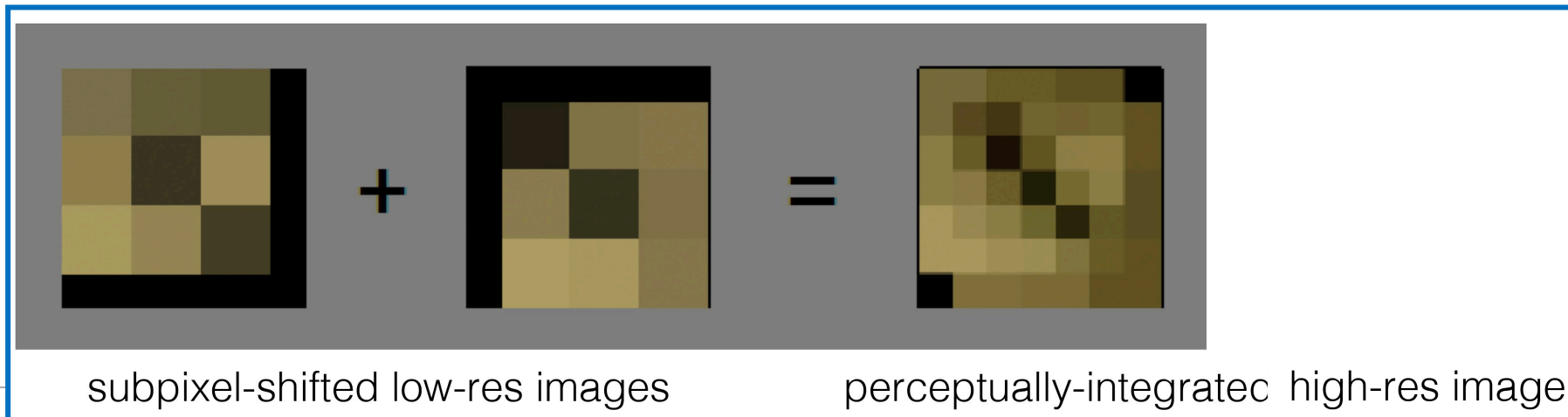
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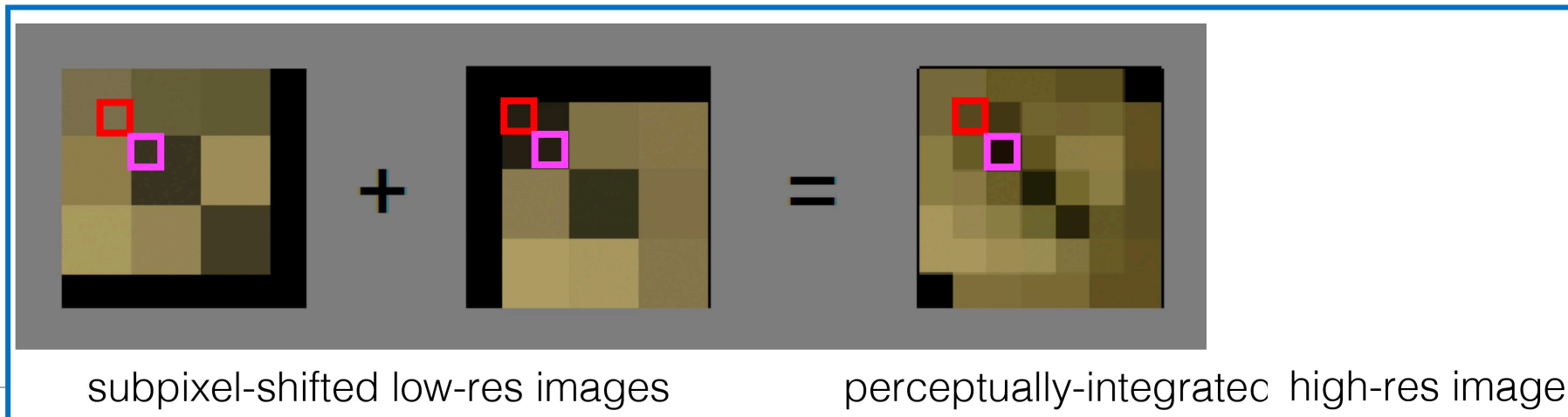
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Increasing spatial resolution

Trivial answer: Pack more pixels!

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- Optical superposition:

- Projection principle.
- Low-resolution images from multiple devices are optically superimposed on a projection screen.
- Superimposed images are shifted by some amount with respect to each other => one super-resolved pixel receives contributions from multiple devices.
- Precise calibration is essential.

- Temporal superposition

- Similar to optical, but with temporal multiplexing.
- Wobulation, vibrating displays, apparent resolution increase...

Spatial resolution – Wobulation (Temp. superposition)

- **Wobulation:** Technique to increase spatial resolution in projection displays. How does it work?



unwobulated



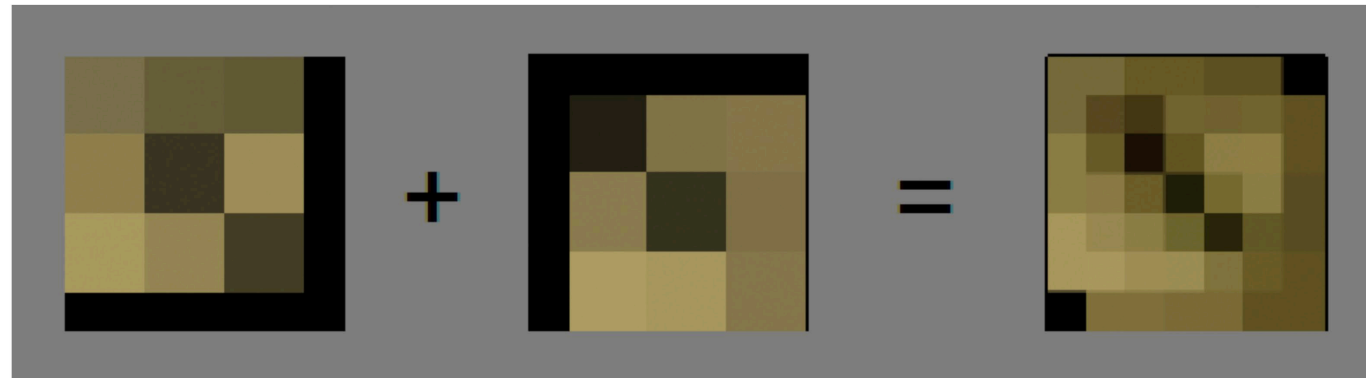
wobulated

[Allen and Ulichney 2005]

Spatial resolution – Wobulation (Temp. superposition)

- **Wobulation:** Technique to increase spatial resolution in projection displays.
How does it work?

- For each frame, multiple sub-frames are generated
- Each sub-frame is projected onto the screen in a slightly different position => pixels from distinct sub-frames overlap.
- Sub-frames are projected in fast* succession similar to individual frames in a movie.
- This rapid sequence is perceived as a continuous, flicker-free image.
- Detail in the resulting image is increased.



subpixel-shifted low-res images

perceptually-integrated

high-res image

* Faster than the perceivable flickering frequency of the HVS.

Spatial resolution – Wobulation (Temp. superposition)



A

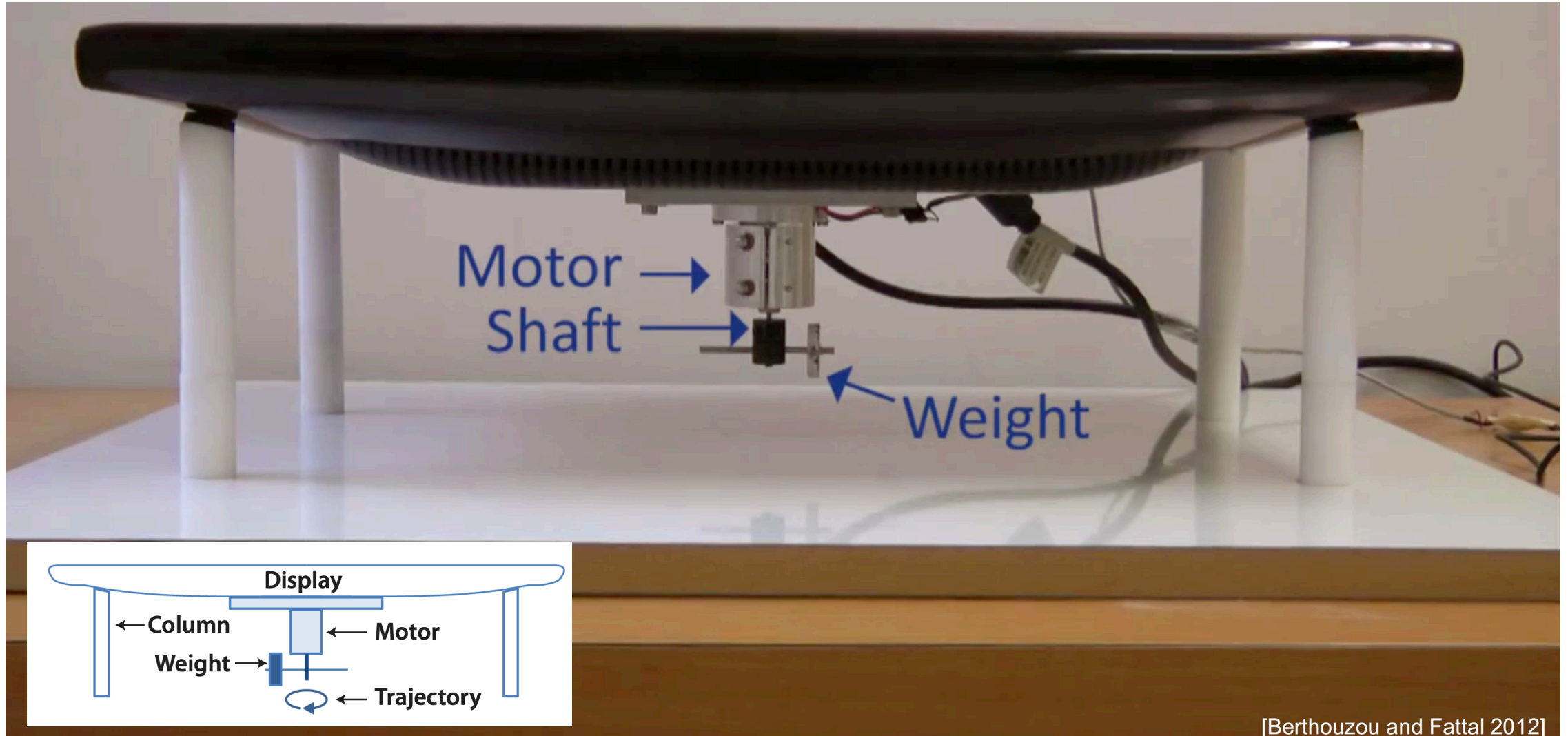
A + B

(A) Single sub-frame

(A + B) 2 sub-frames
superimposed

Can be more than 2 sub-frames (but requires higher speed switching)

Spatial resolution – Vibrating displays (Temp. superpos.)



[Berthouzou and Fattal 2012] Resolution enhancement by vibrating displays. TOG.

Temporal superresolution displays

Perceptual considerations – Hold-type blur

- HVS is limited in perceiving high temporal frequencies
 - Non-instantaneous response of retina receptors & high-level vision processes
- Temporal variations of signal => averaged and perceived as constant signal
- **Critical flicker frequency (CFF)**: Threshold frequency for a signal to be perceived as constant or as varying over time.
 - Value depends on: luminance adaptation, spatial extent of stimulus, retinal region...
- Consequence of this temporal averaging: **hold-type blur**

Hold-type blur

- **In real world:** Objects move continuously and are stabilized in retina and well tracked by the HVS => enabled by smooth pursuit eye motion (SPEM).
- **In current displays:** Our HVS tracks the images, but images are static for an extended period of time (the period of a frame).

Hold-type blur

- **In real world:** Objects move continuously and are stabilized in retina and well tracked by the HVS => enabled by smooth pursuit eye motion (SPEM).
- **In current displays:** Our HVS tracks the images, but images are static for an extended period of time (the period of a frame).
- Due to temporal averaging, receptors on the retina average the signal while moving across the image during the period of one frame.
- As a result, the perceived image is blurred => **hold-type blur**, purely perceptual (cannot be observed in the content!).

Increasing temporal resolution

Trivial answer: Use higher framerate

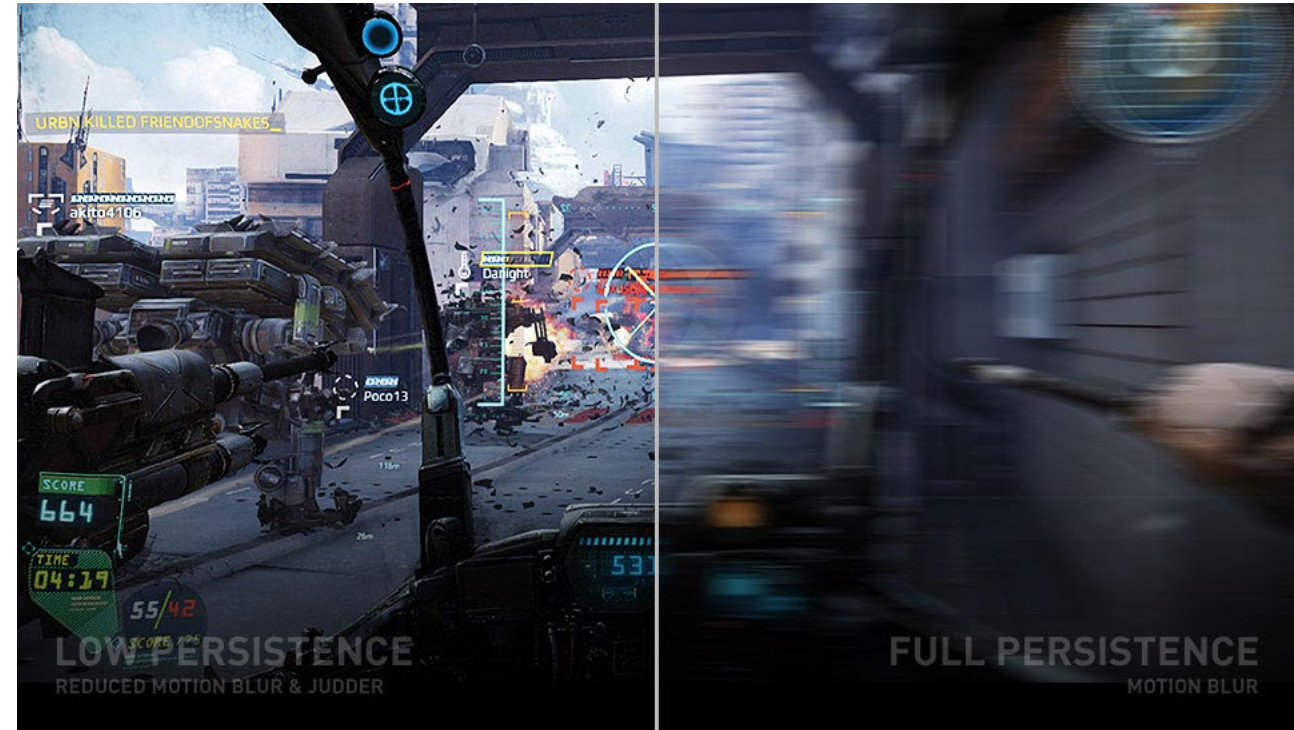
Besides this, different techniques:

Increasing temporal resolution

Trivial answer: Use higher framerate

Besides this, different techniques:

- **Black data insertion:**
 - Introduces black frames interleaved with the original content.
 - Reduces hold-type blur (because it reduces the time during which objects are shown in the same position).
 - It can produce temporal flickering and reduced brightness.



Low persistence displays interleave black frames with original content to reduce hold-type blur.

Increasing temporal resolution

Trivial answer: Use higher framerate

Besides this, different techniques:

- **Frame interpolation:**

- Interpolate original frames along motion trajectories (requires optical flow!)
- [+] Can, e.g., go from 24Hz to 240Hz wo/ brightness reduction or flickering.
- [–] Optical flow artifacts, efficiency, lag and suitability for interactive scenarios.



Input Image A



Interpolated Frames

Input Image B

Increasing temporal resolution

Impulse-Type displays: Briefly flash each frame on the screen and then show black until the next frame (e.g., CRTs or some modern gaming monitors). Similar to black data insertion.

Drawbacks:

- Flicker sensitivity if low refresh rates
- Reduced brightness (because the screen is not continuously illuminated)

Stereo and light field displays

Why stereo?

- Improves realism
- Images are no longer flat
- Can reproduce view-dependent effects => e.g., better material perception



Brief history of stereo

1838: different images for both eyes

1890: patent on 3D movies

1900: tripod for taking 3D pictures

1915: exhibition of 3D images

1922: 3D movie

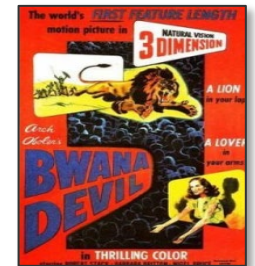
1923: 3D movie with stereo sound

1952: 3D movie in color

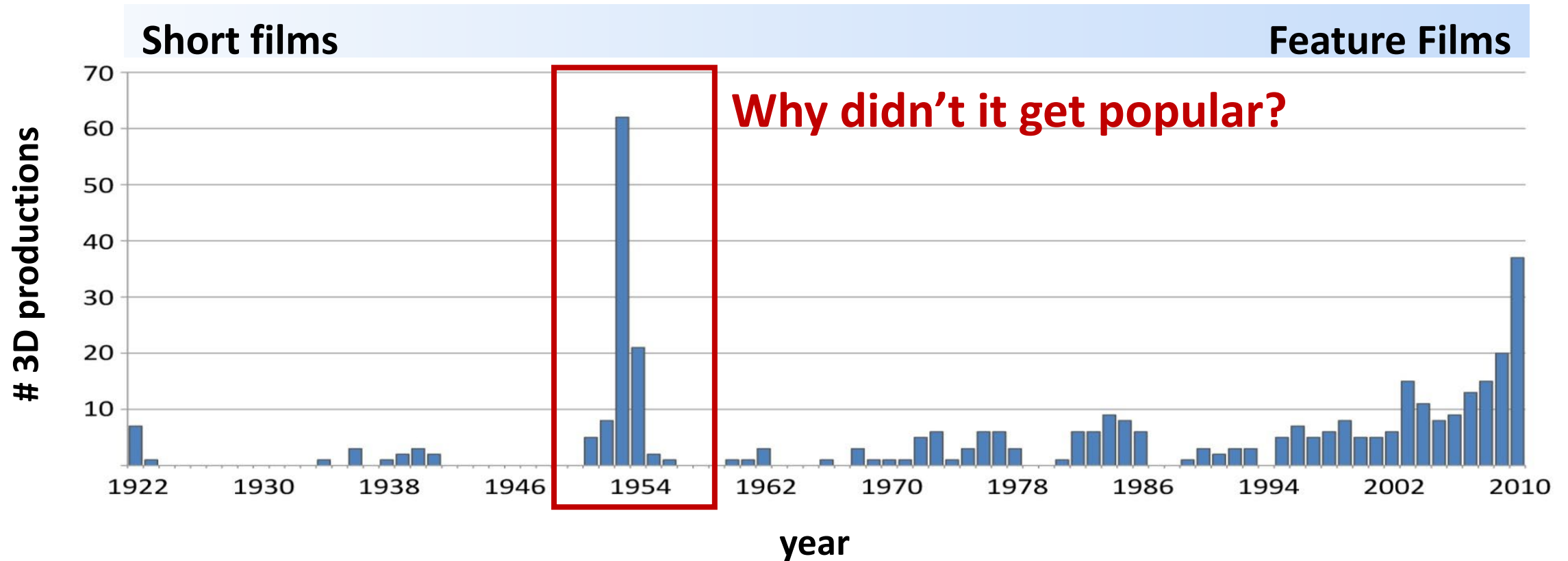
1990s: IMAX cinemas, TV series

2003: feature film in 3D for IMAX

2009 - now: increasing popularity



Number of 3D productions (1922-2010)

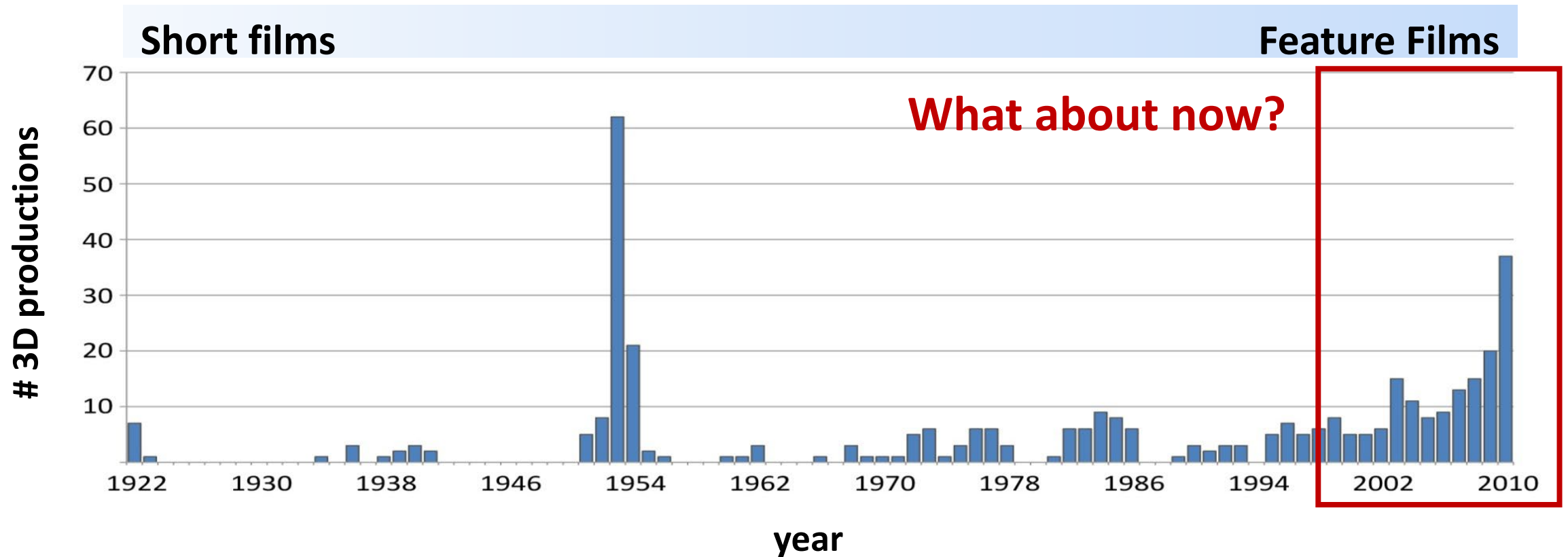


Early stereo productions

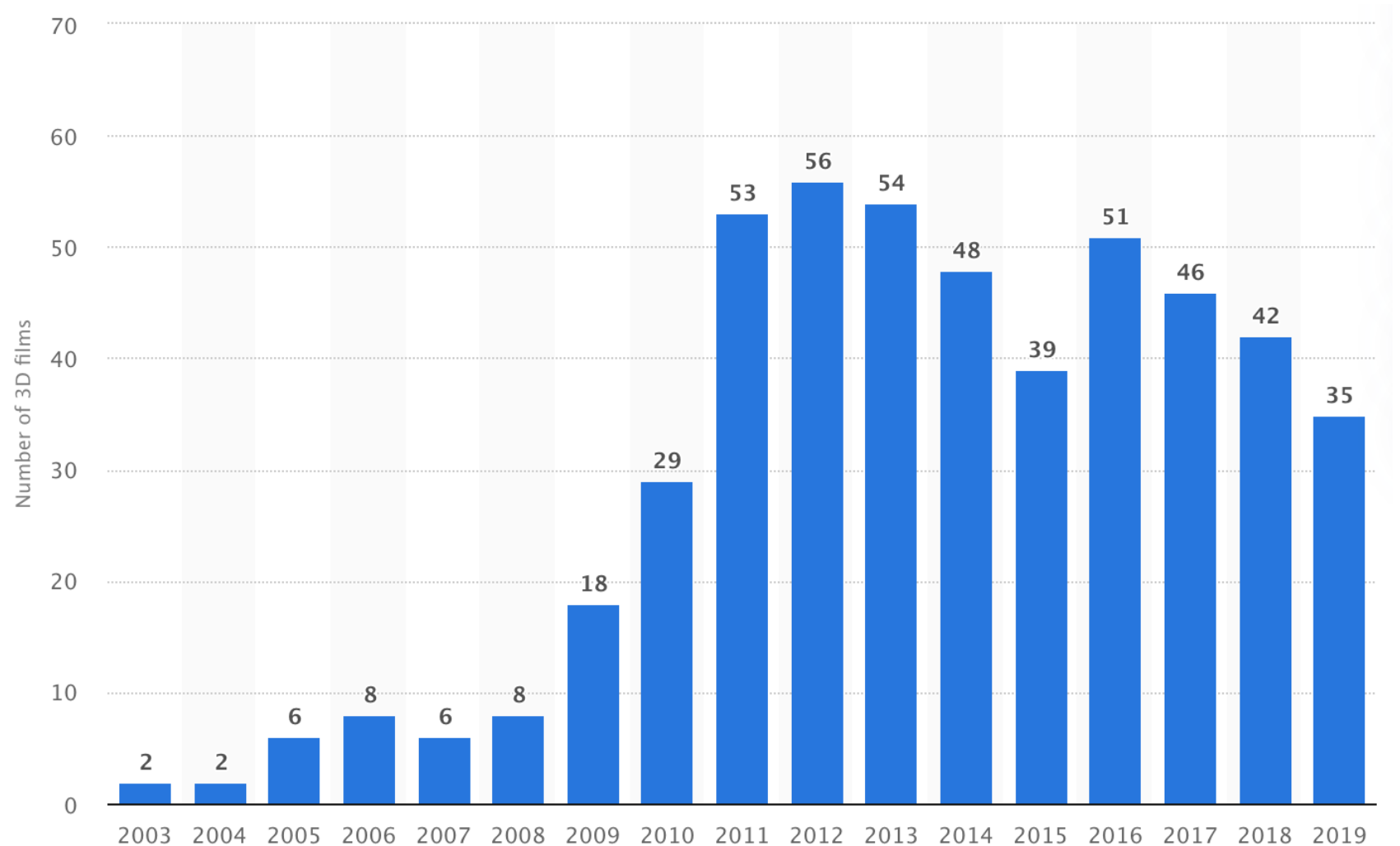
- Expensive hardware
- Lack of standardized format
- Impossible at home
- Lack of interesting content



Number of 3D productions (1922-2010)



Number of 3D productions (2003-2019, US & Canada)

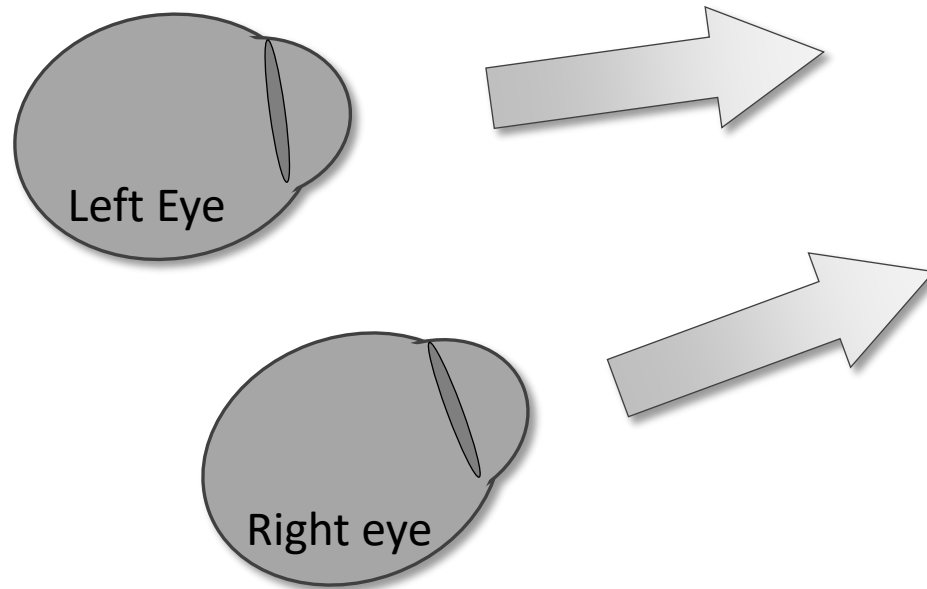


Current stereo production

- Great content
 - Beautiful shots with complex depth
 - Computer-generated special effects
 - Still heavily relying on experience of creator
 - Common in theaters & VR
 - Equipment is less expensive
 - But not really pervasive at home
 - Virtual reality
 - Equipment is much improved
 - But still: missing cues & conflict problems
- They are flat!*

How is stereo vision simulated on a flat display?

- Provide a different image to each eye

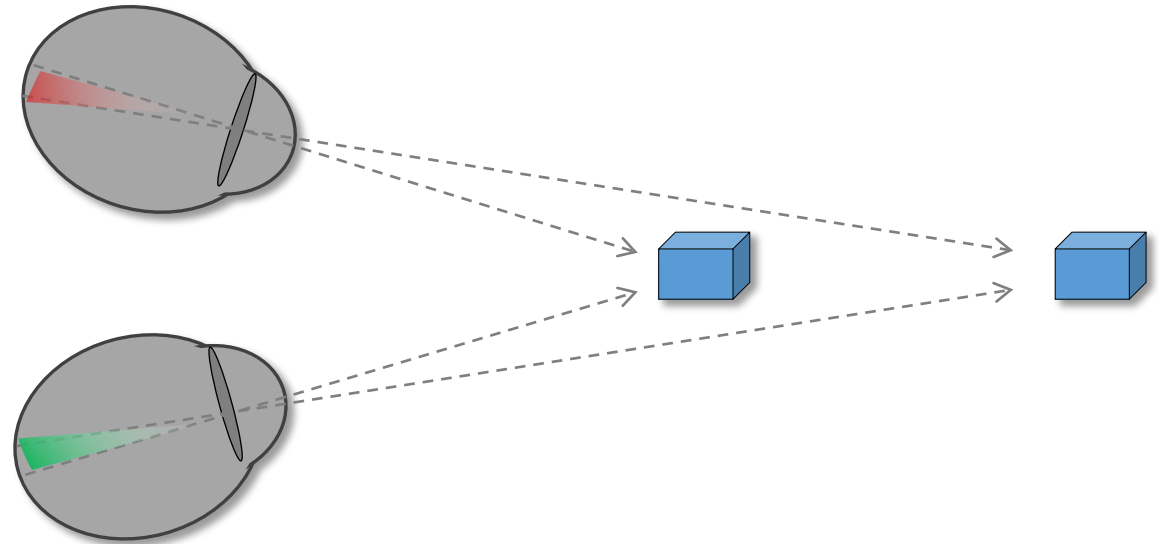


Depth perception

- We perceive depth thanks to a number of depth cues.

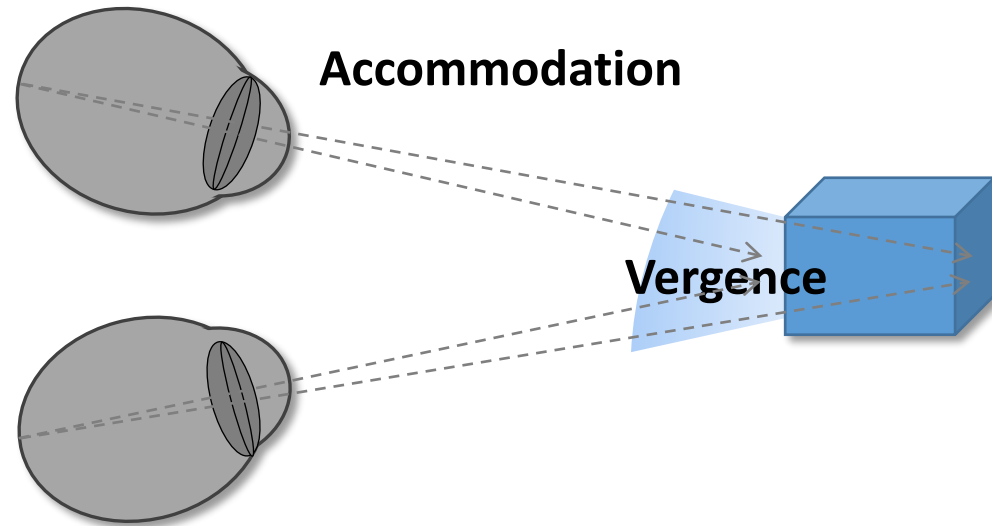
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- Stereoscopic depth cues:
binocular depth cues



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- Stereoscopic depth cues:
binocular depth cues
- Ocular depth cues:
accommodation, vergence



Depth perception

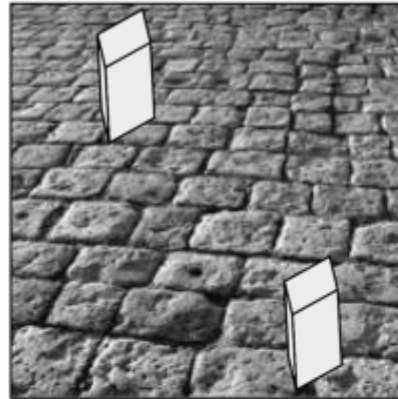
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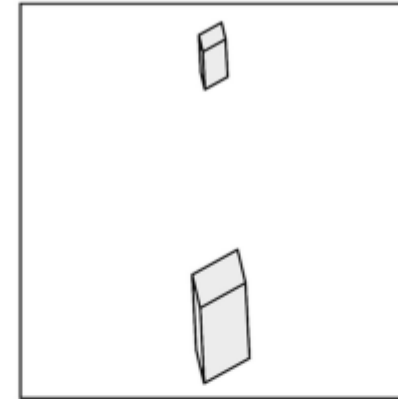
- Ocular depth cues:
accommodation, vergence

- Pictorial depth cues:
occlusion, size, shadows...

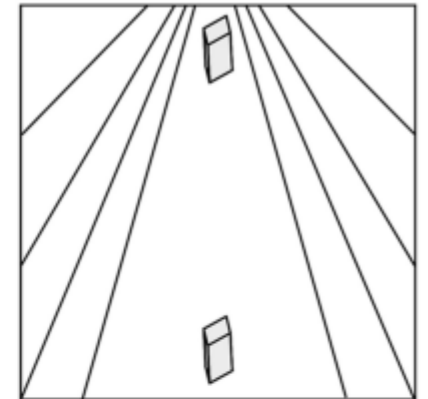
Texture Gradient



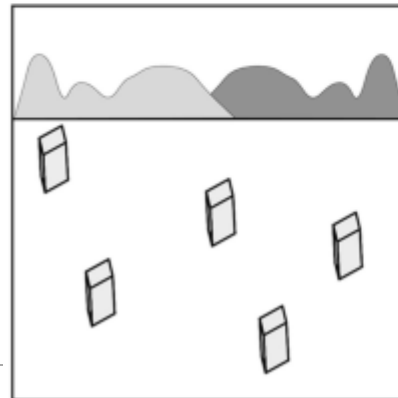
Relative Size



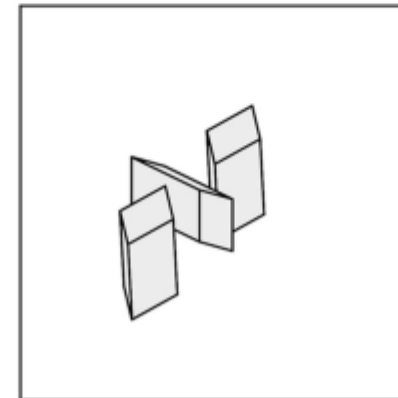
Linear Perspective



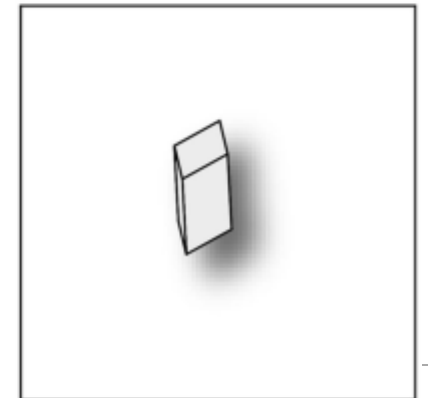
Position Relative To Horizon



Occlusion



Shadows



Source: Dantone et al. Multidimensional Image Retargeting Course.

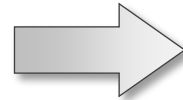
Depth perception – Cues in conflict

- We perceive depth thanks to a number of depth cues.
- Different cues are in accordance in the real world.
- But they can conflict in displays or synthetic images.

Depth perception

- We perceive depth thanks to a number of depth cues.

- Stereoscopic depth cues:
binocular depth cues
- Ocular depth cues:
accommodation, vergence

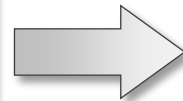


Require 3D space

We cheat our visual system!

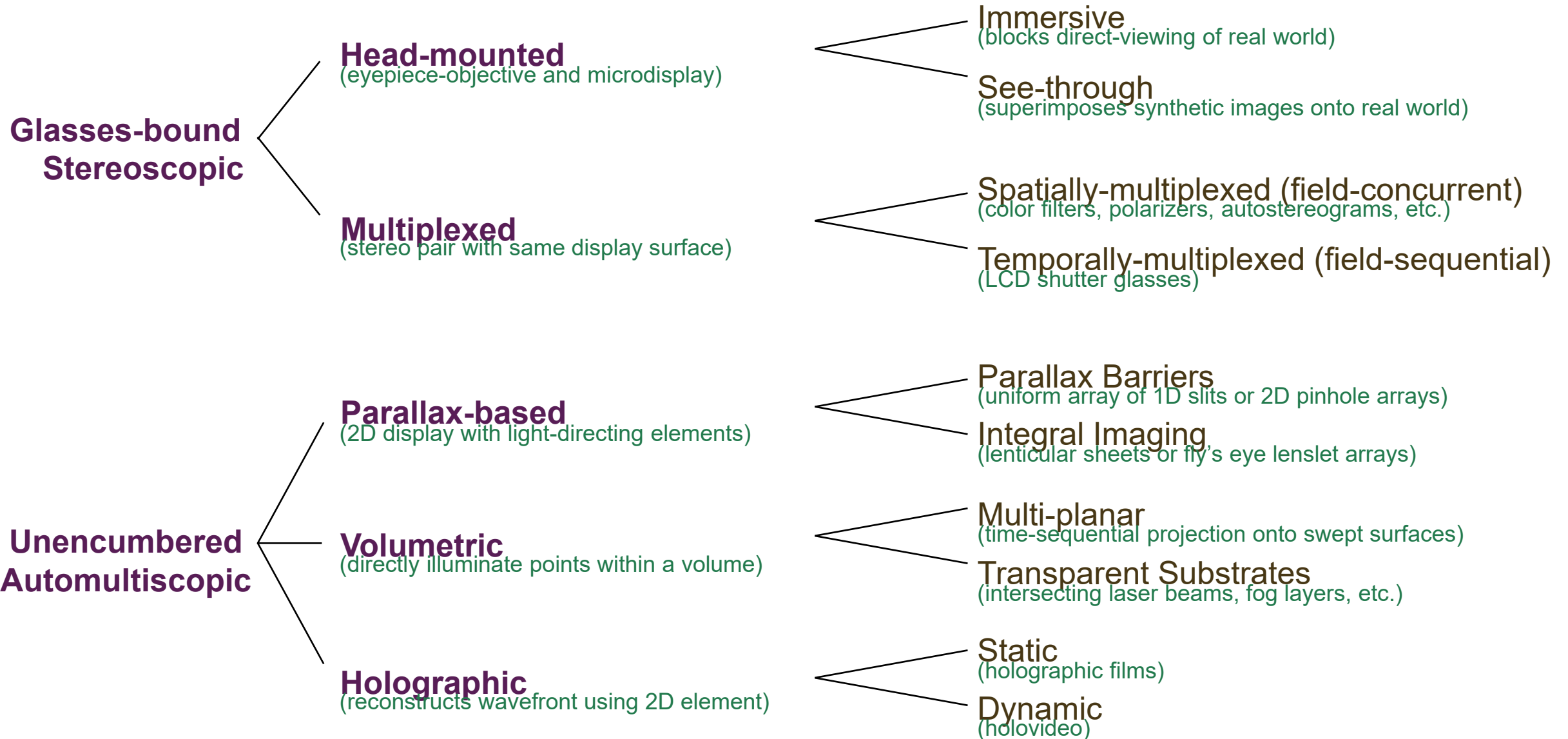
3D displays attempt to reproduce these.

- Pictorial depth cues:
occlusion, size, shadows...

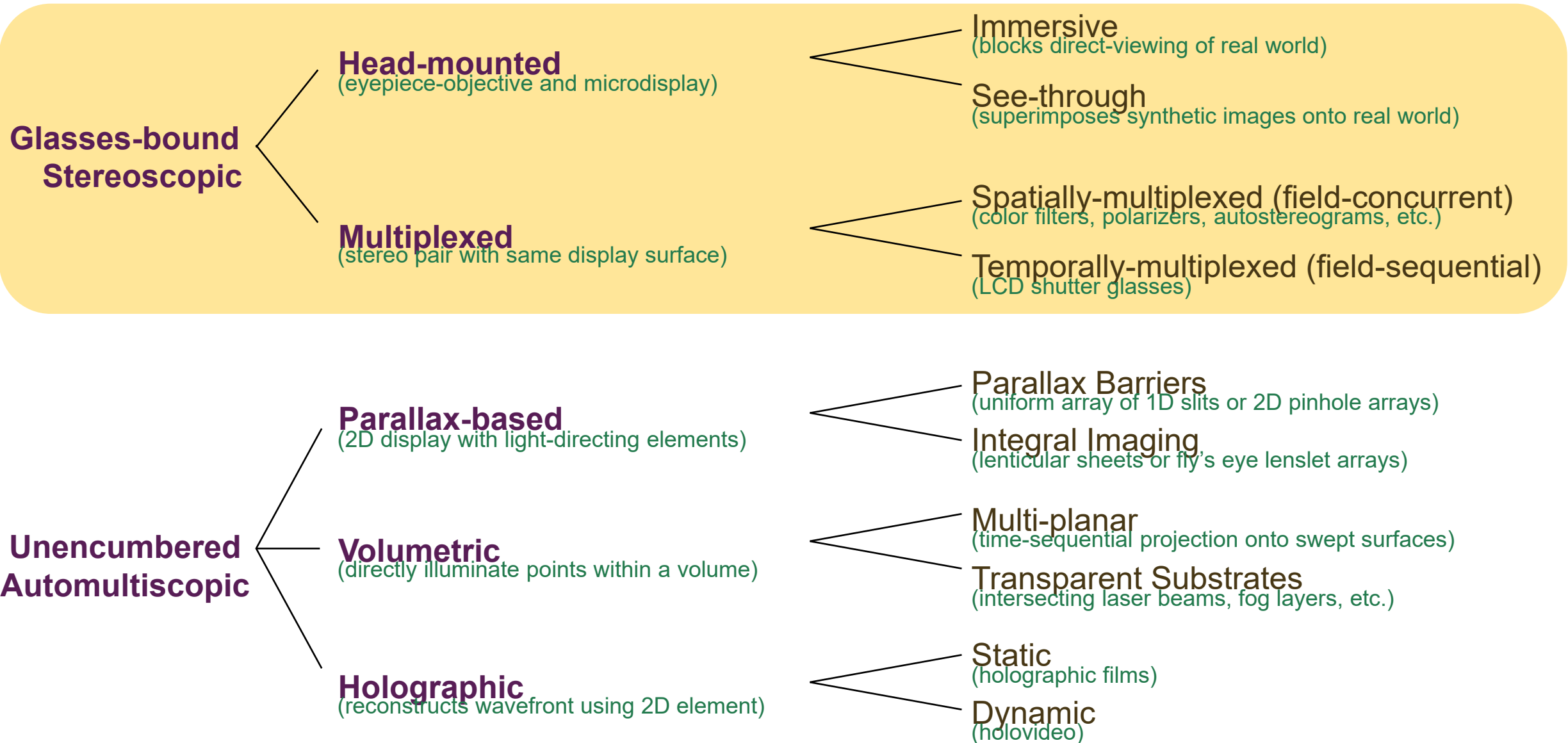


Reproducible on flat displays

A Taxonomy of 3D Displays



A Taxonomy of 3D Displays



Immersive Head-Mounted Displays (HMDs)



**Glasses-bound
Stereoscopic**

Head-mounted
(eyepiece-objective and microdisplay)

Multiplexed
(stereo pair with same display surface)

Immersive
(blocks direct-viewing of real world)

See-through Head-Mounted Displays (HMDs)



**Glasses-bound
Stereoscopic**

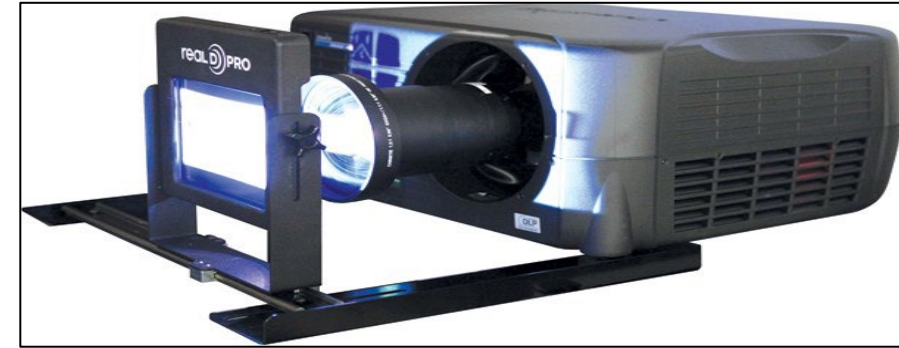
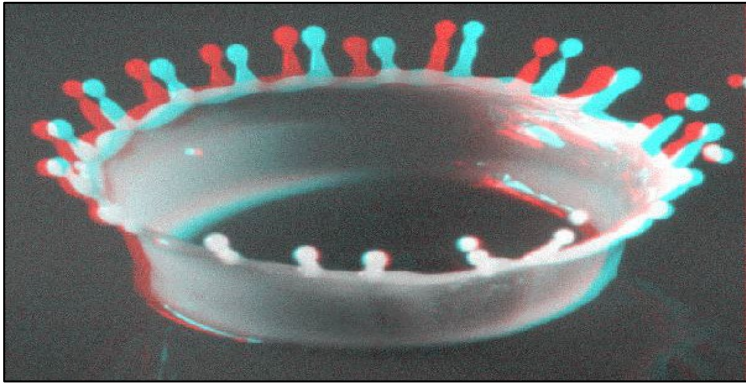
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Spatially-multiplexed stereo displays



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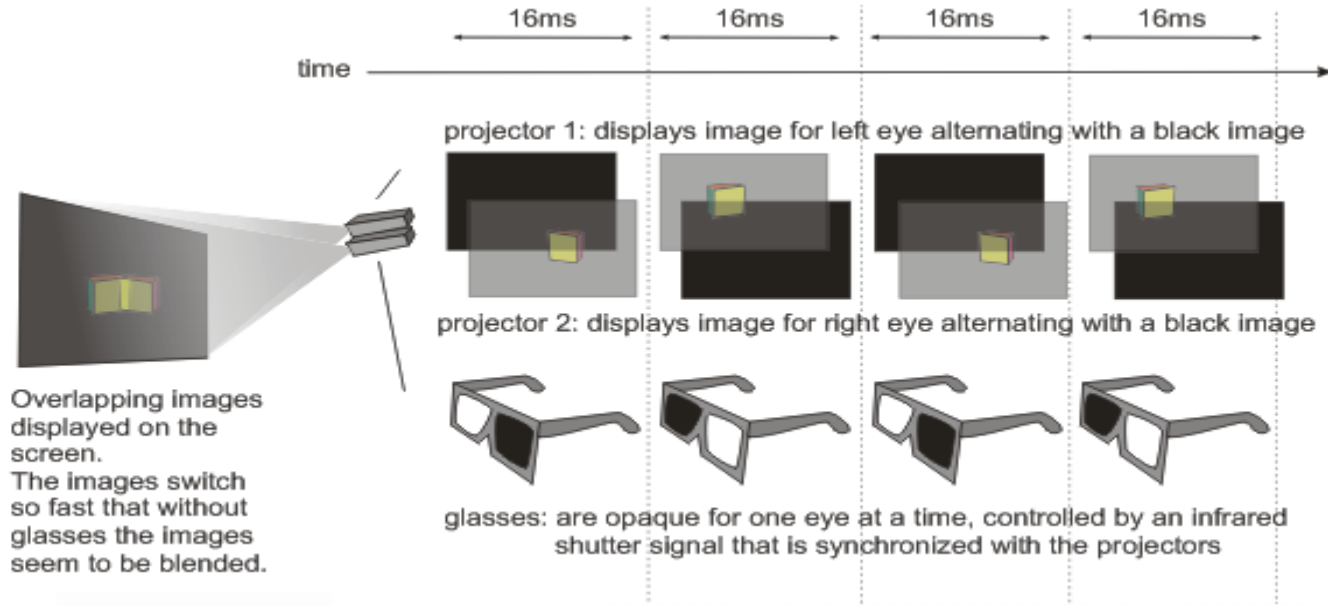
Immersive
(blocks direct-viewing of real world)

See-through
(superimposes synthetic images onto real world)

Spatially-multiplexed (field-concurrent)
(color filters, polarizers, etc.)

Temporally-multiplexed (field-sequential)
(LCD shutter glasses)

Temporally-multiplexed stereo displays



Glasses-bound
Stereoscopic

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(stereo pair with same display surface)

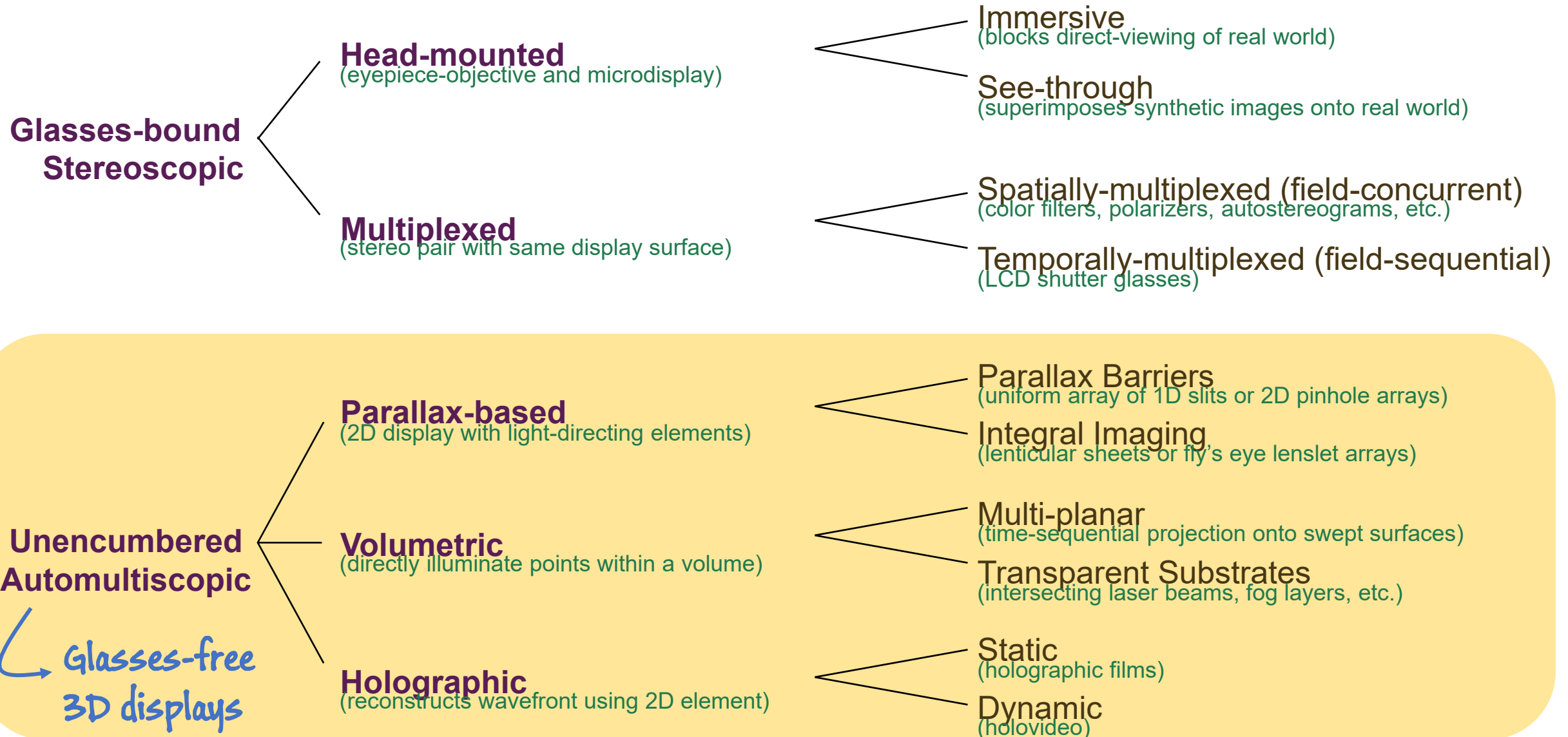
Immersive
(blocks direct-viewing of real world)

See-through
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Spatially-multiplexed (field-concurrent)
(color filters, polarizers, autostereograms, etc.)

Temporally-multiplexed (field-sequential)
(LCD shutter glasses)

A Taxonomy of 3D Displays



Are “glasses-free” 3D displays ready?

Today?



Nintendo 3DS
E3 2010



MasterImage 3D
Computex 2011



Asus Eee Pad Memo 3D
Computex 2011



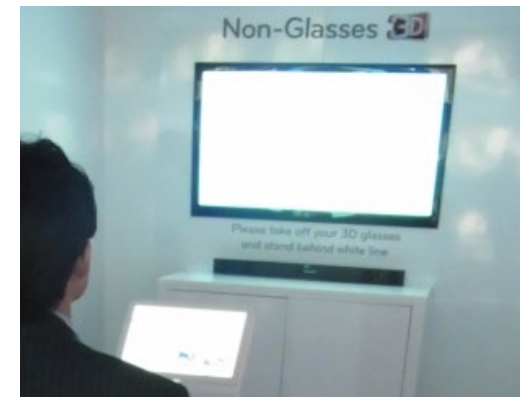
LG Optimus 3D
Mobile World Congress 2011



Toshiba 3DTV Prototype
CES 2011



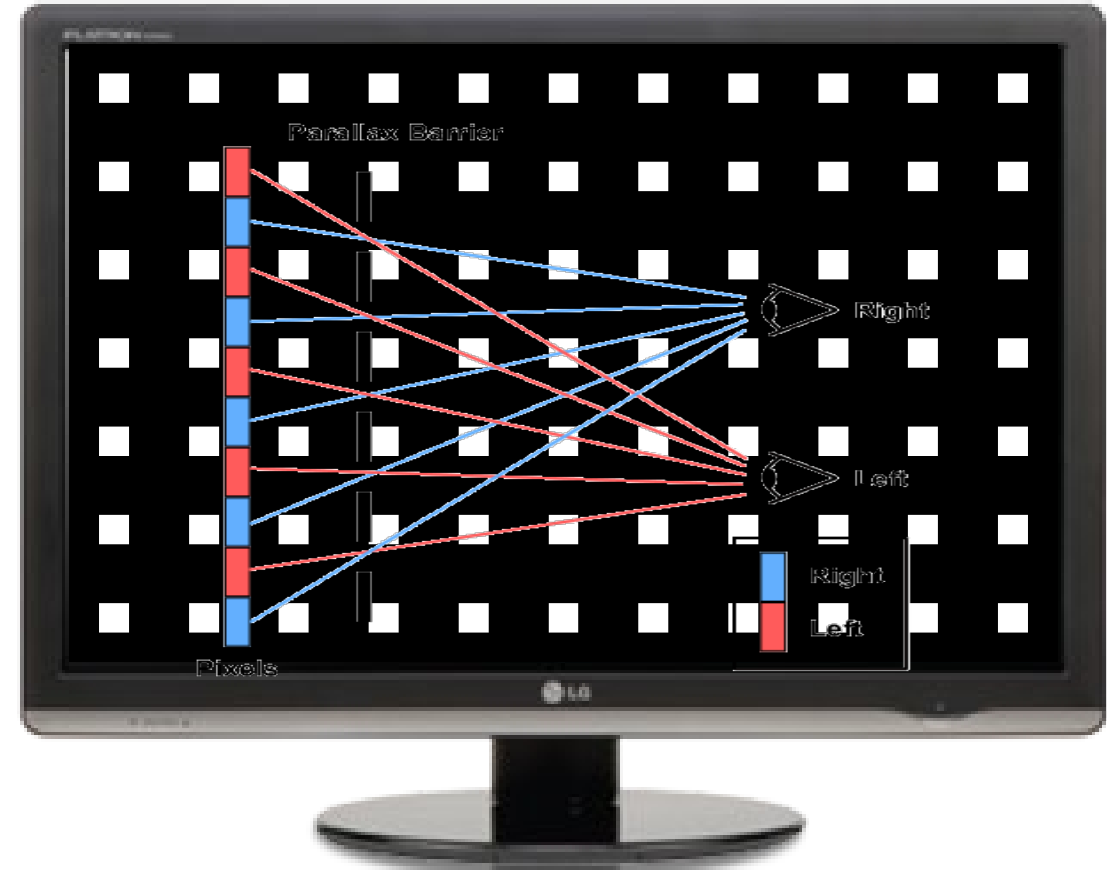
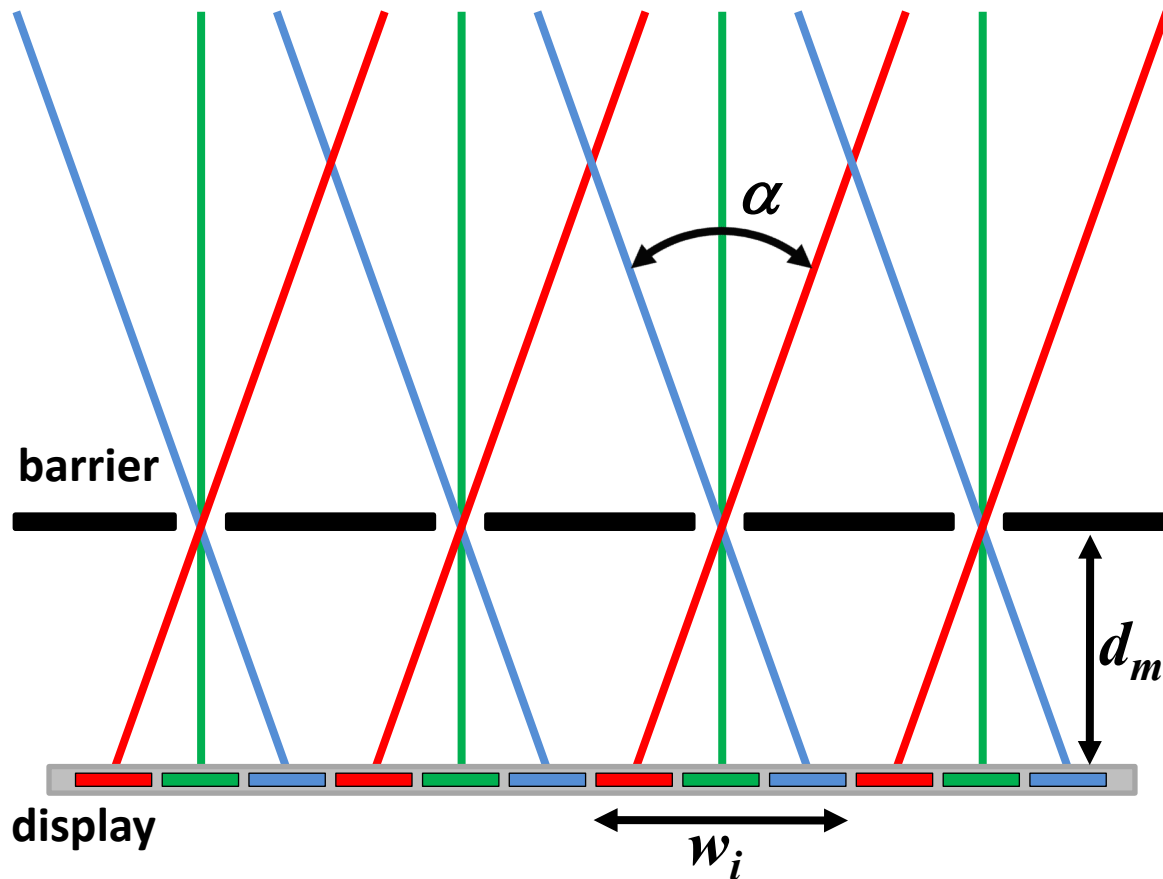
Sony 3DTV Prototype
CES 2011



LG 3DTV Prototype
CES 2011

Glasses-free 3D displays – Parallax barriers

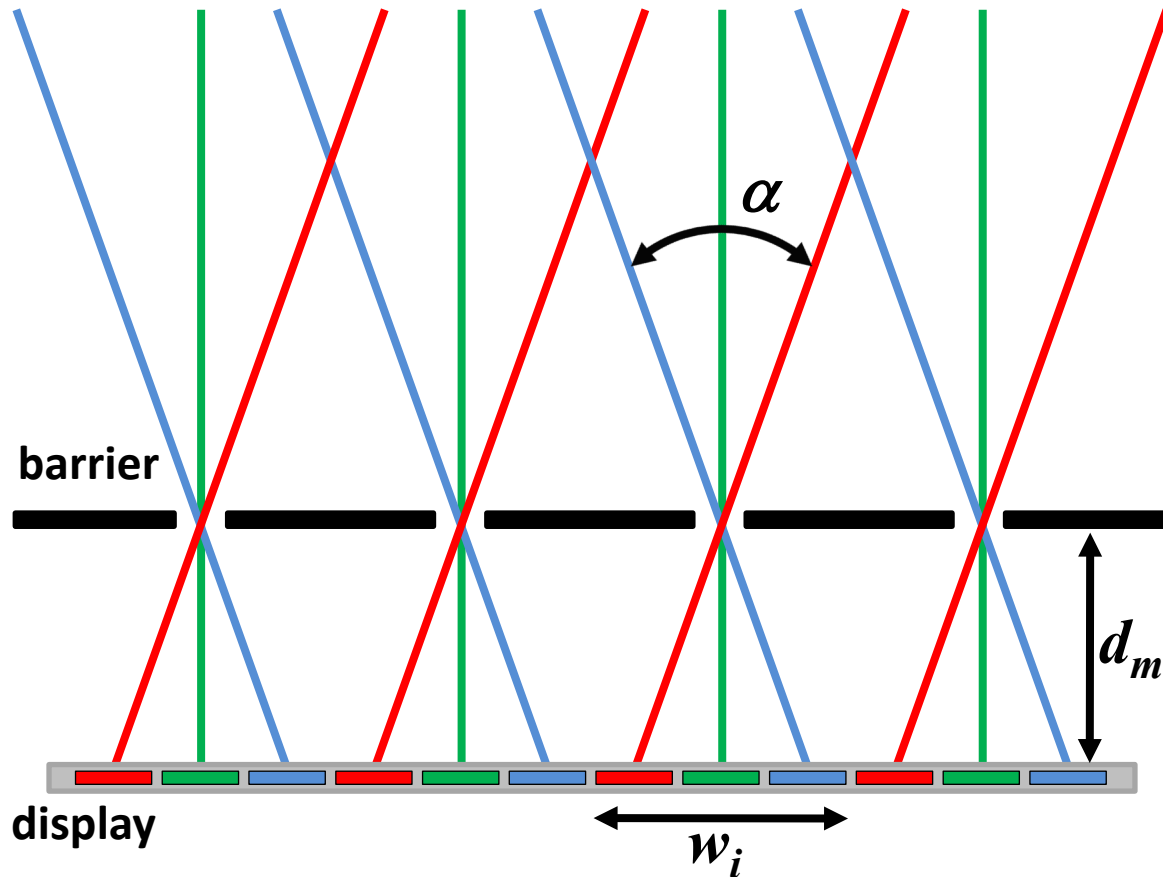
It may remind you of something...



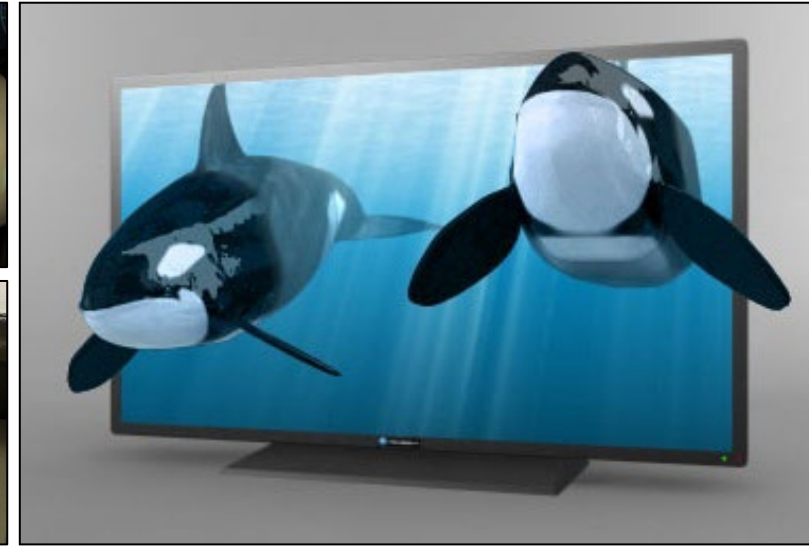
- Introduced by Frederic E. Ives in 1903
- Requires light-attenuating layers (i.e., masks)
- Allows multi-view display, but results in dim images due to attenuation

Glasses-free 3D displays – Parallax barriers

It may remind you of something...



Range Rover 2010
Dual-View Navigation

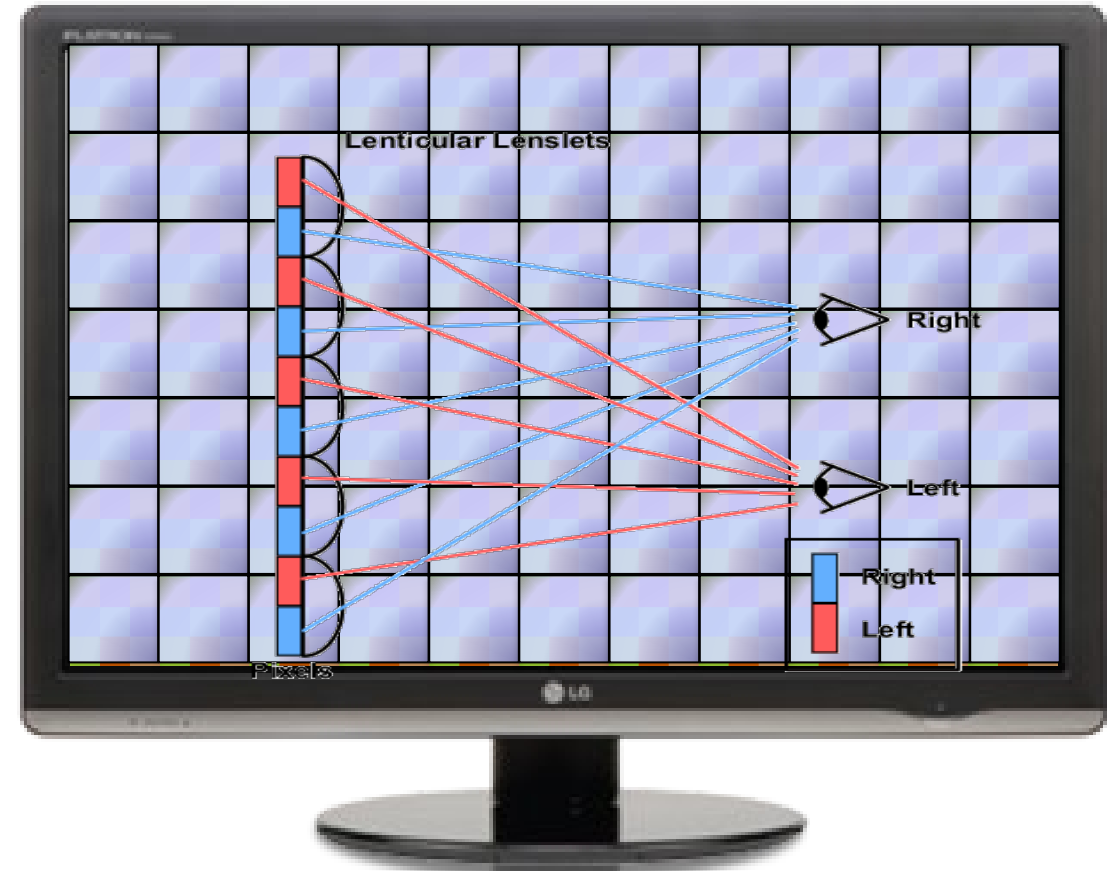
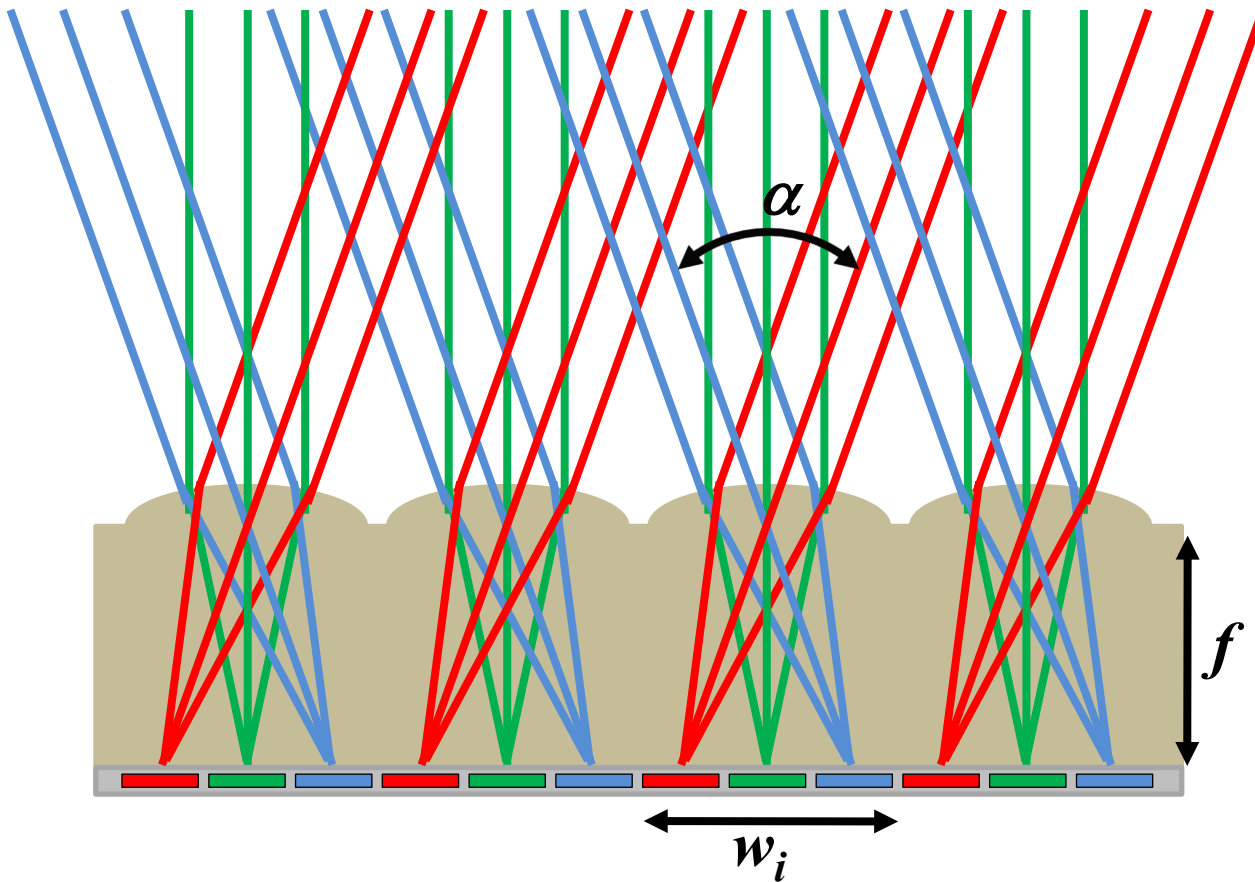


NewSight MV-42AD3 42"
(1920x1080, 1x8 views)

- Introduced by Frederic E. Ives in 1903
- Requires two light-attenuating layers (i.e., masks)
- Allows multi-view display, but results in dim images due to attenuation

Glasses-free 3D displays – Integral imaging

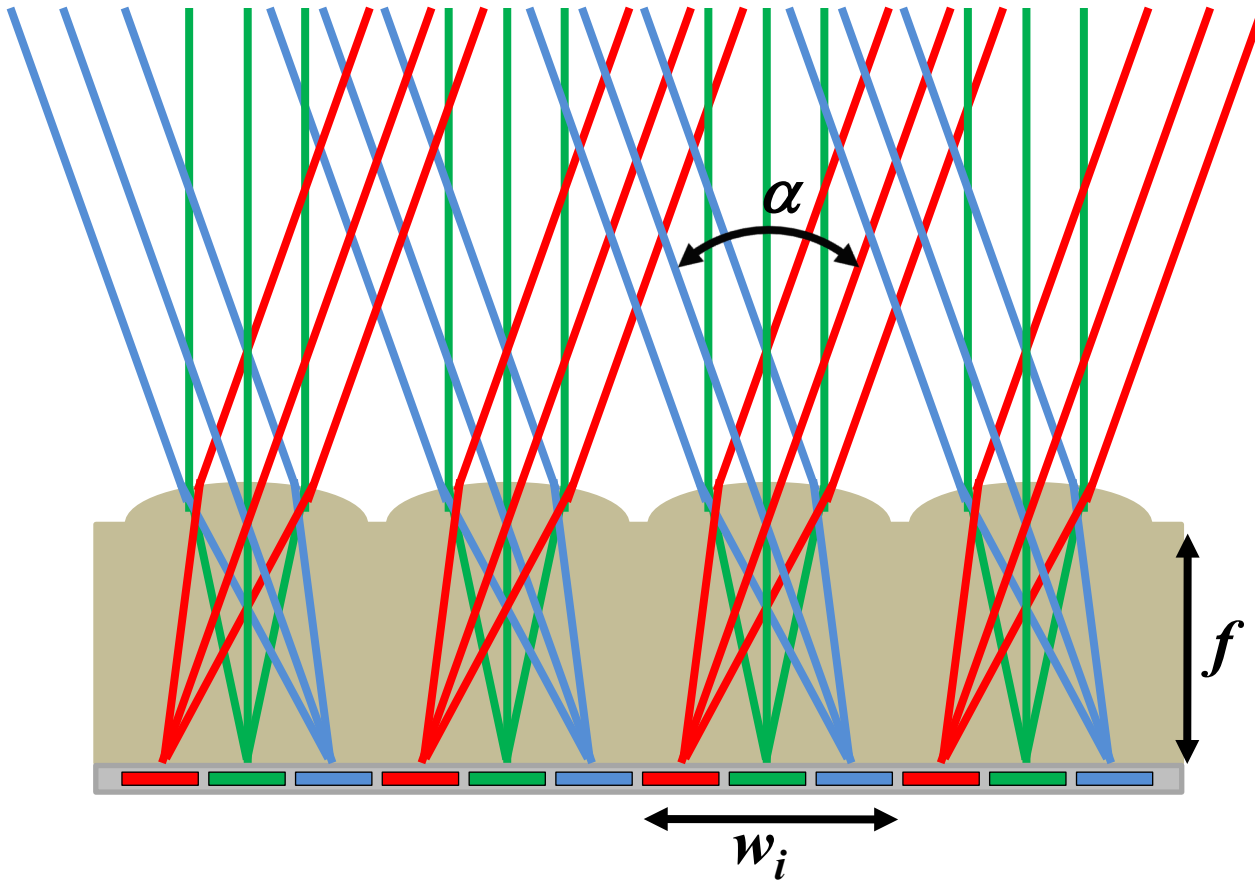
It may remind you of something...



- Introduced by Gabriel Lippmann in 1908
- Requires a refractive lenslet array to be affixed to a conventional display
- Allows multi-view display, but imposes fixed trade-off between spatial and angular resolution

Glasses-free 3D displays – Integral imaging

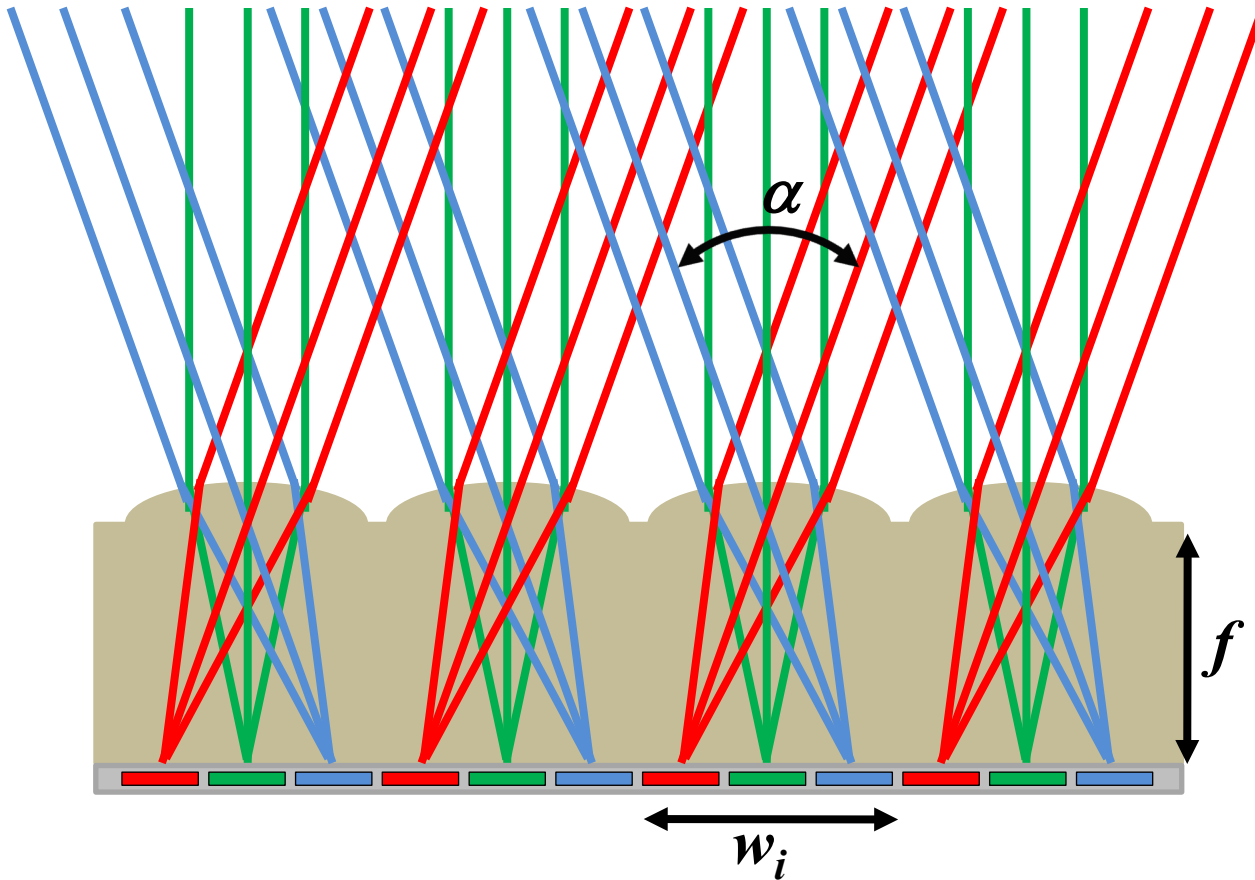
It may remind you of something...



Alioscopy 3DHD 42"
(1920x1200, 1x8 views)

- Introduced by Gabriel Lippmann in 1908
- Requires a refractive lenslet array to be affixed to a conventional display
- Allows multi-view display, but imposes fixed trade-off between spatial and angular resolution

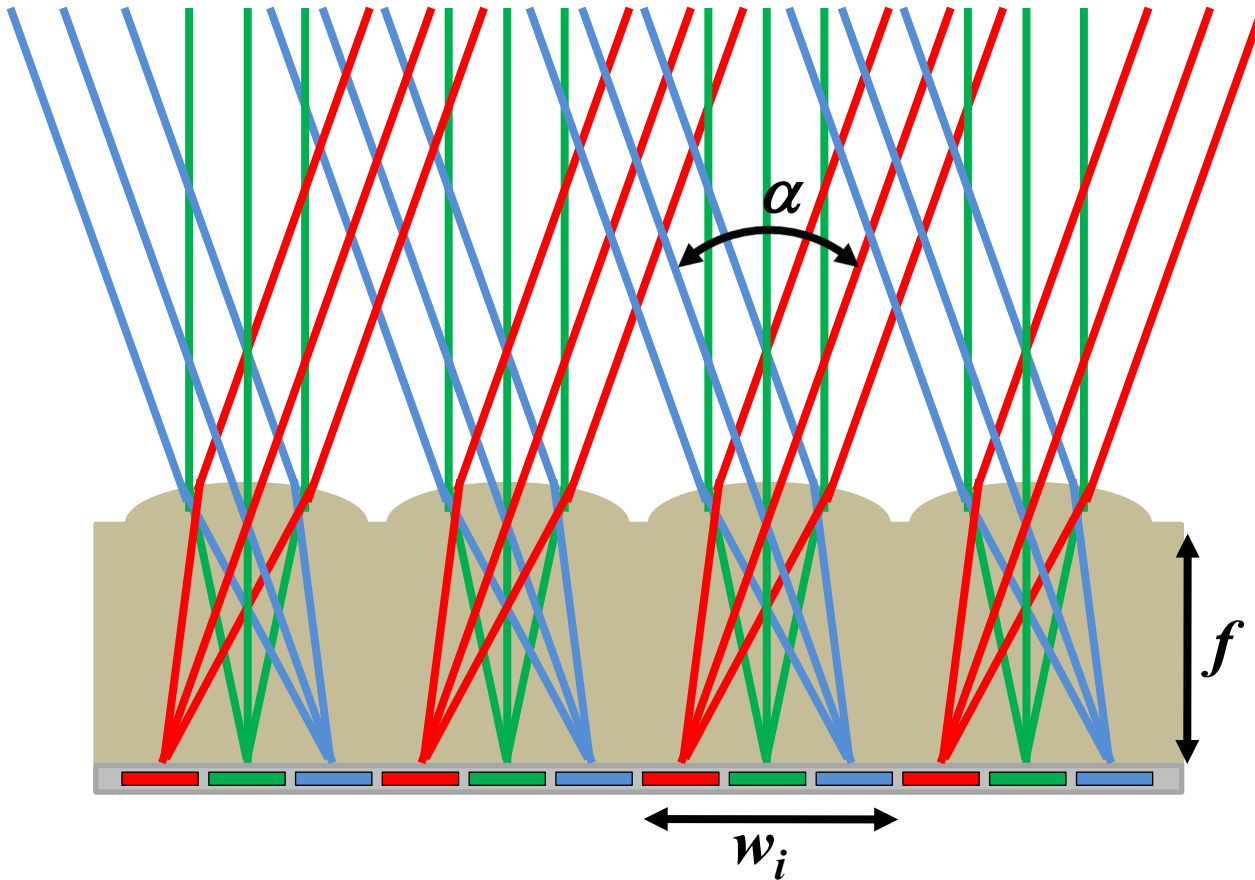
Glasses-free 3D displays – Integral imaging



Does this remind you of anything?

- Introduced by Gabriel Lippmann in 1908
- Requires a refractive lenslet array to be affixed to a conventional display
- Allows multi-view display, but imposes fixed trade-off between spatial and angular resolution

Glasses-free 3D displays – Integral imaging

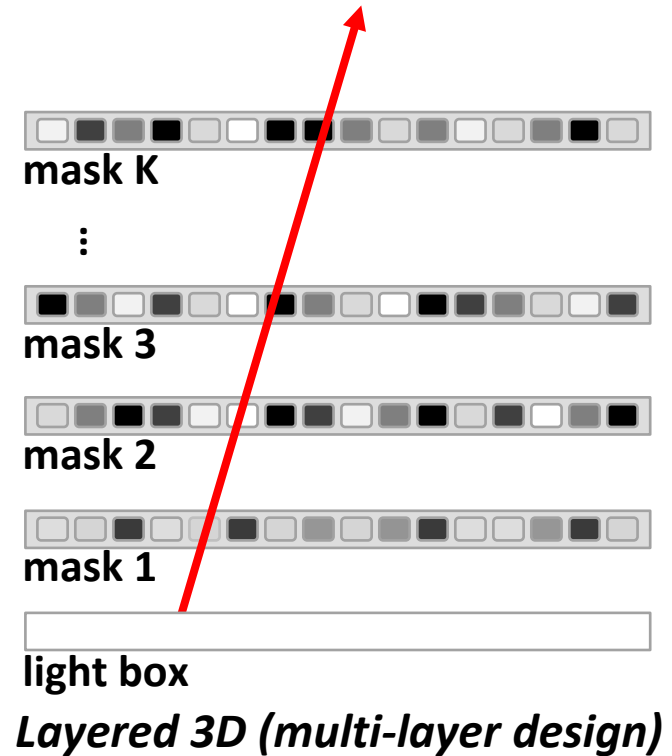
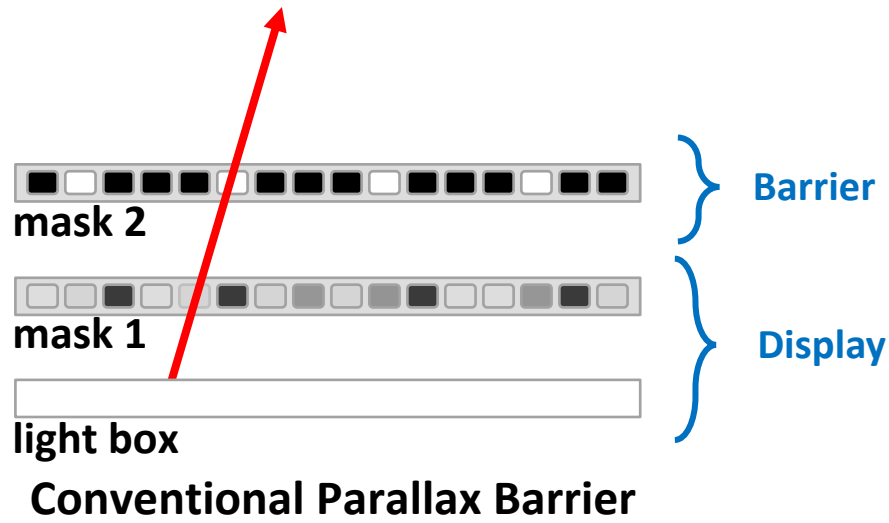


Does this remind you of anything?

They are also termed:
Light field displays

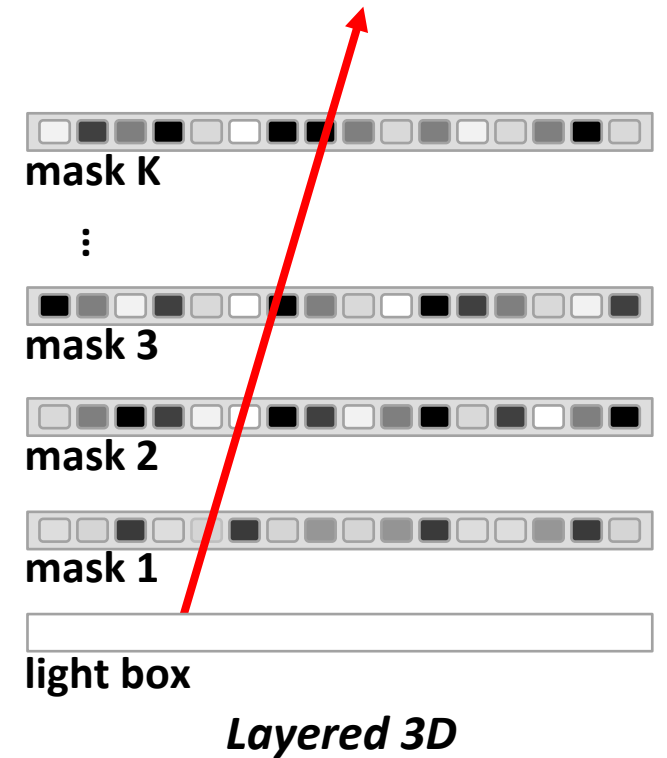
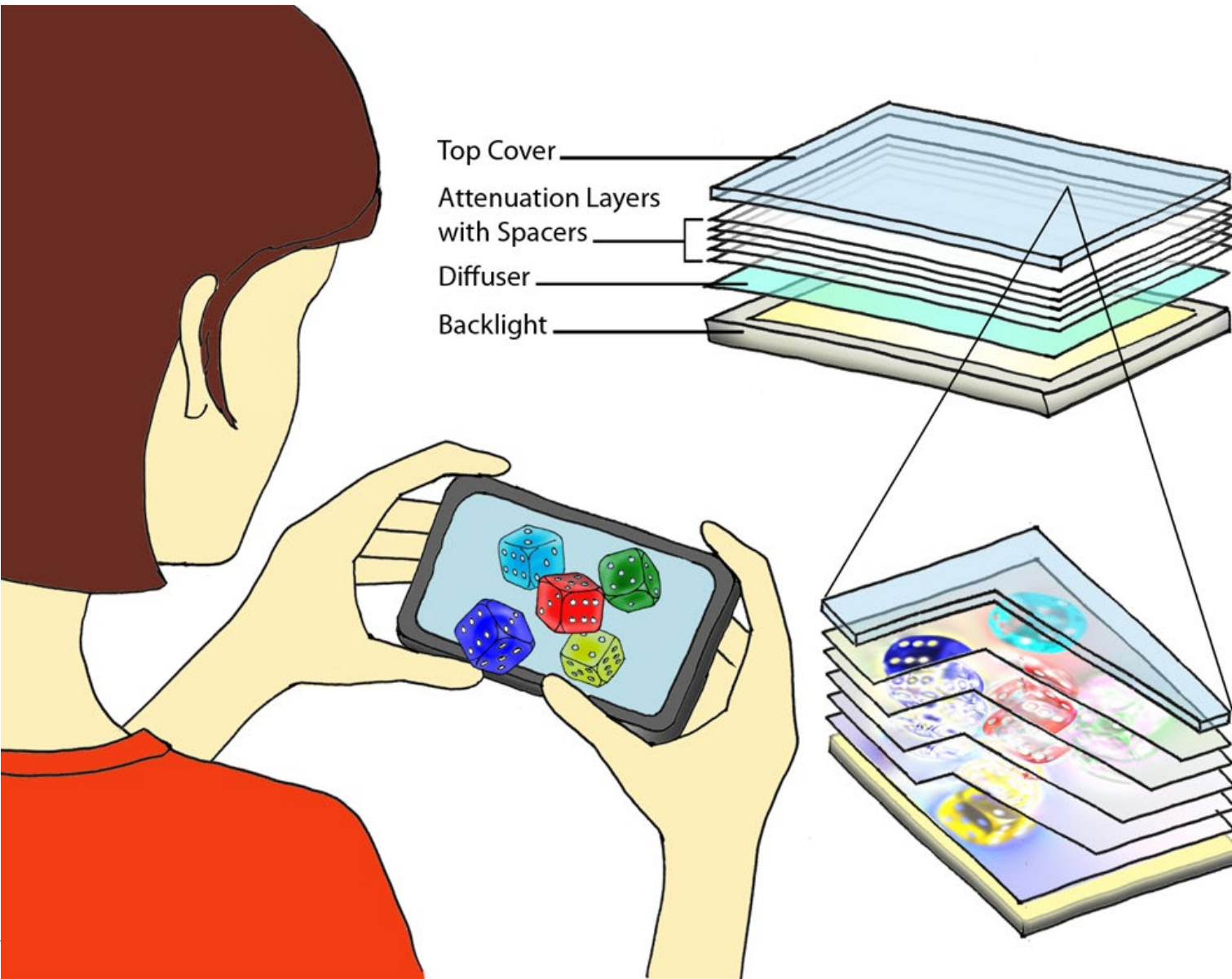
- Introduced by Gabriel Lippmann in 1908
- Requires a refractive lenslet array to be affixed to a conventional display
- Allows multi-view display, but imposes fixed trade-off between spatial and angular resolution

Generalizing Parallax Barriers



- Parallax barriers use *heuristic* design: front mask with slits/pinholes, rear mask with interlaced views
- One could generalize and consider a **multi-layer design** (*Layered 3D*)

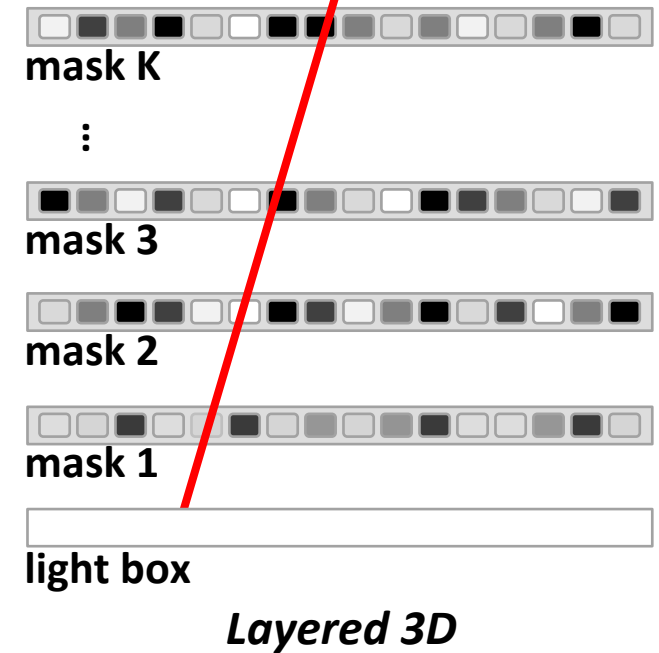
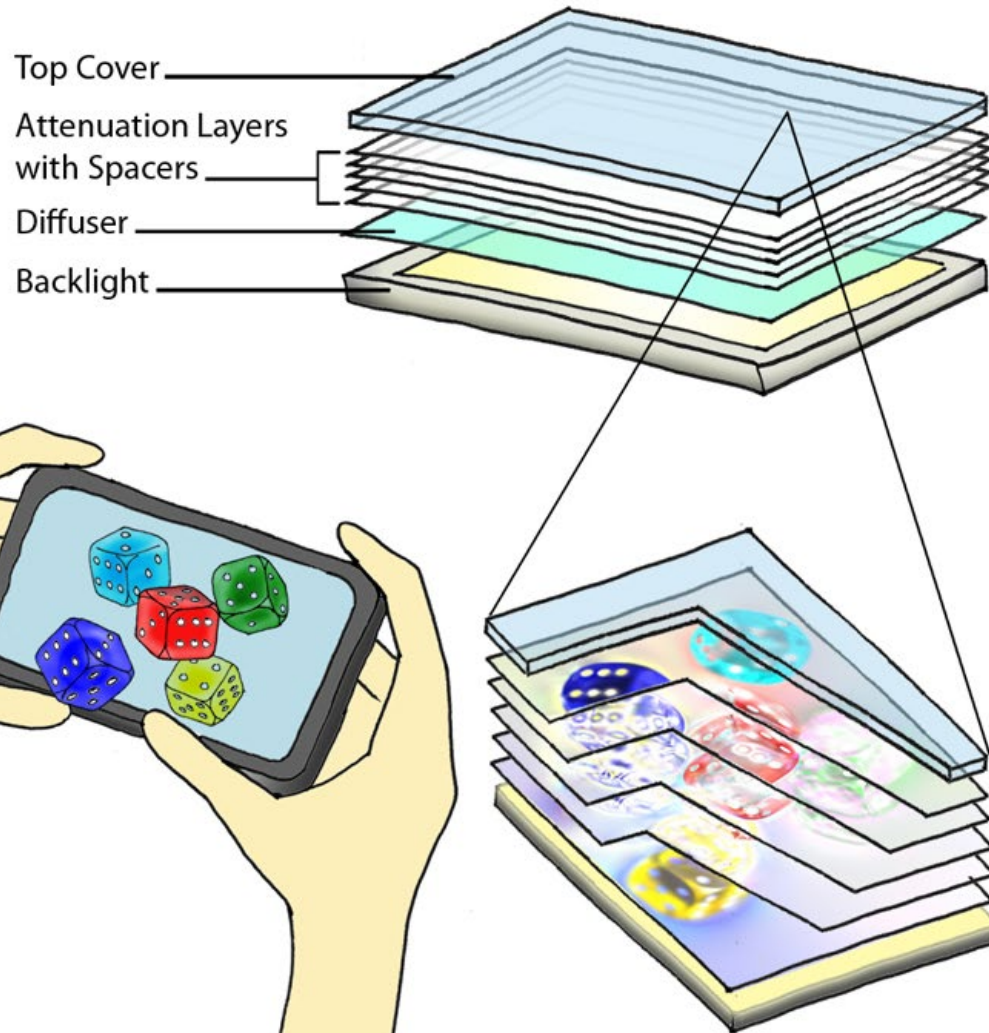
Multi-layer displays: Layered 3D



Multi-layer displays: Layered 3D

The goal is to reproduce a light field of the scene.

But how are these layers computed?

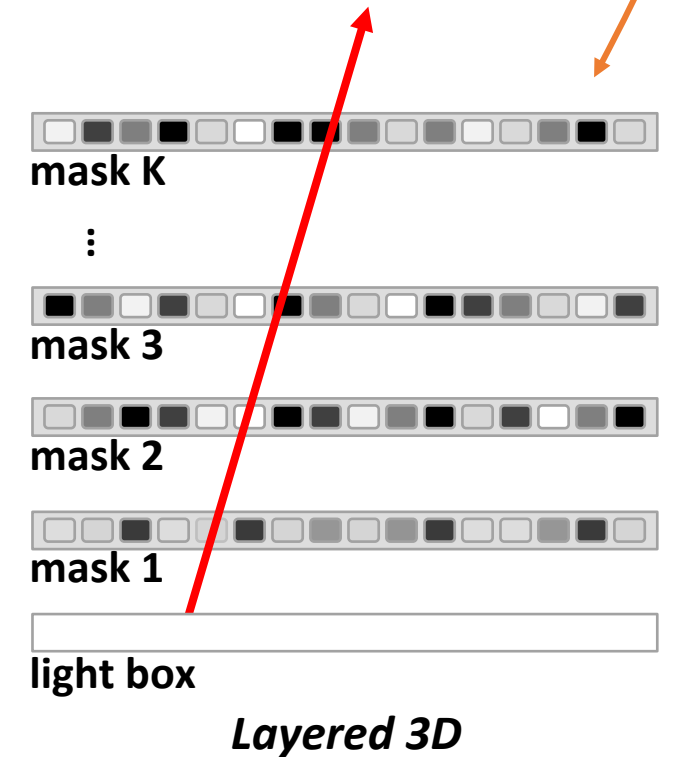


Multi-layer displays: Layered 3D

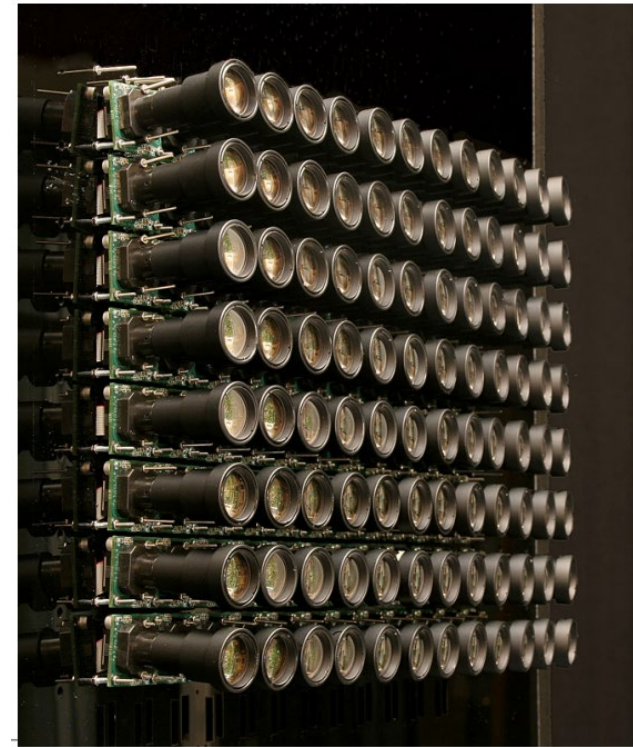
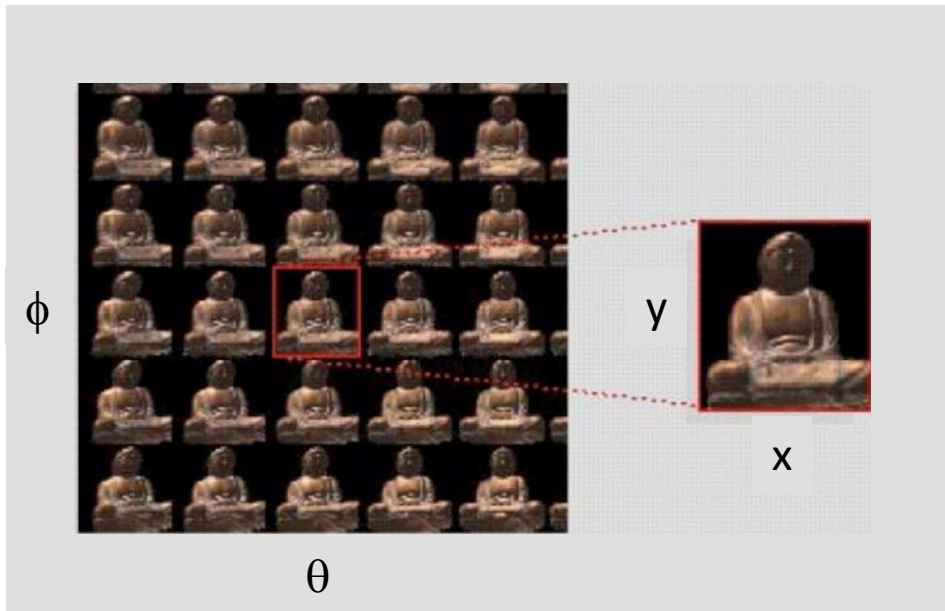
- The goal is to display a light field of the scene (the image viewed changes with viewing angle)
- The way in which the layers are computed builds on **tomography**
- Before diving into it, let's do a brief reminder about light fields...

The goal is to reproduce a light field of the scene.

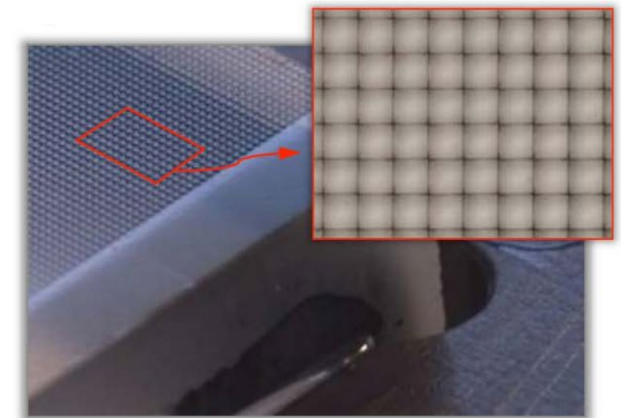
But how are these layers computed?



Light field visualization

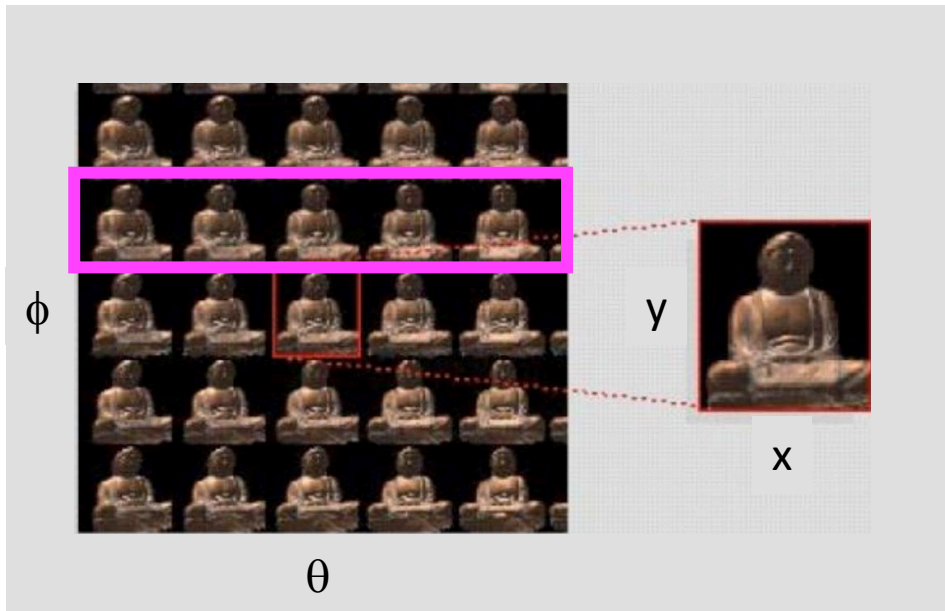


Raytrix™



125μ square-sided microlenses

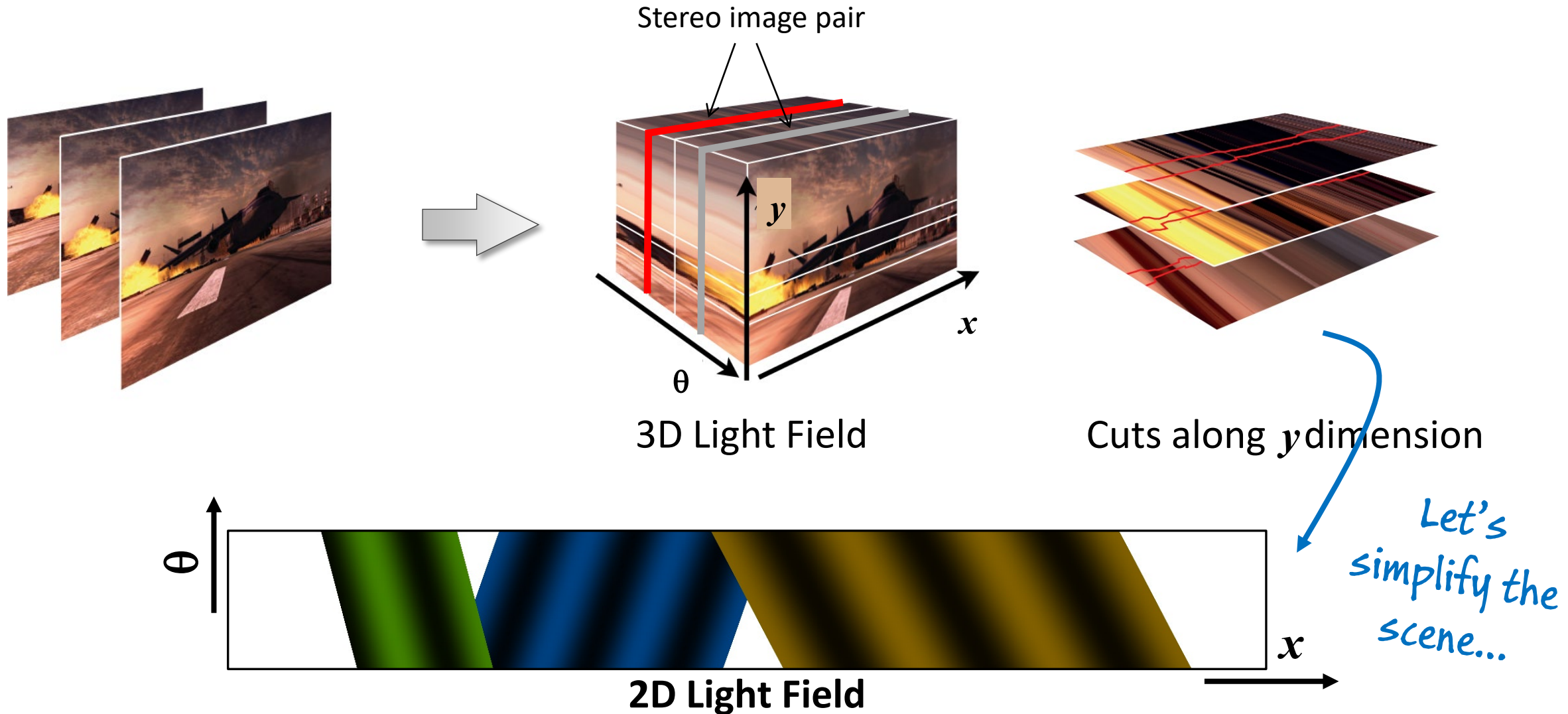
Light field visualization



Let's focus on one row
(horizontal parallax only),
and stack the images from
that row...

(sorry, the buddha is going to
change into an airplane...)

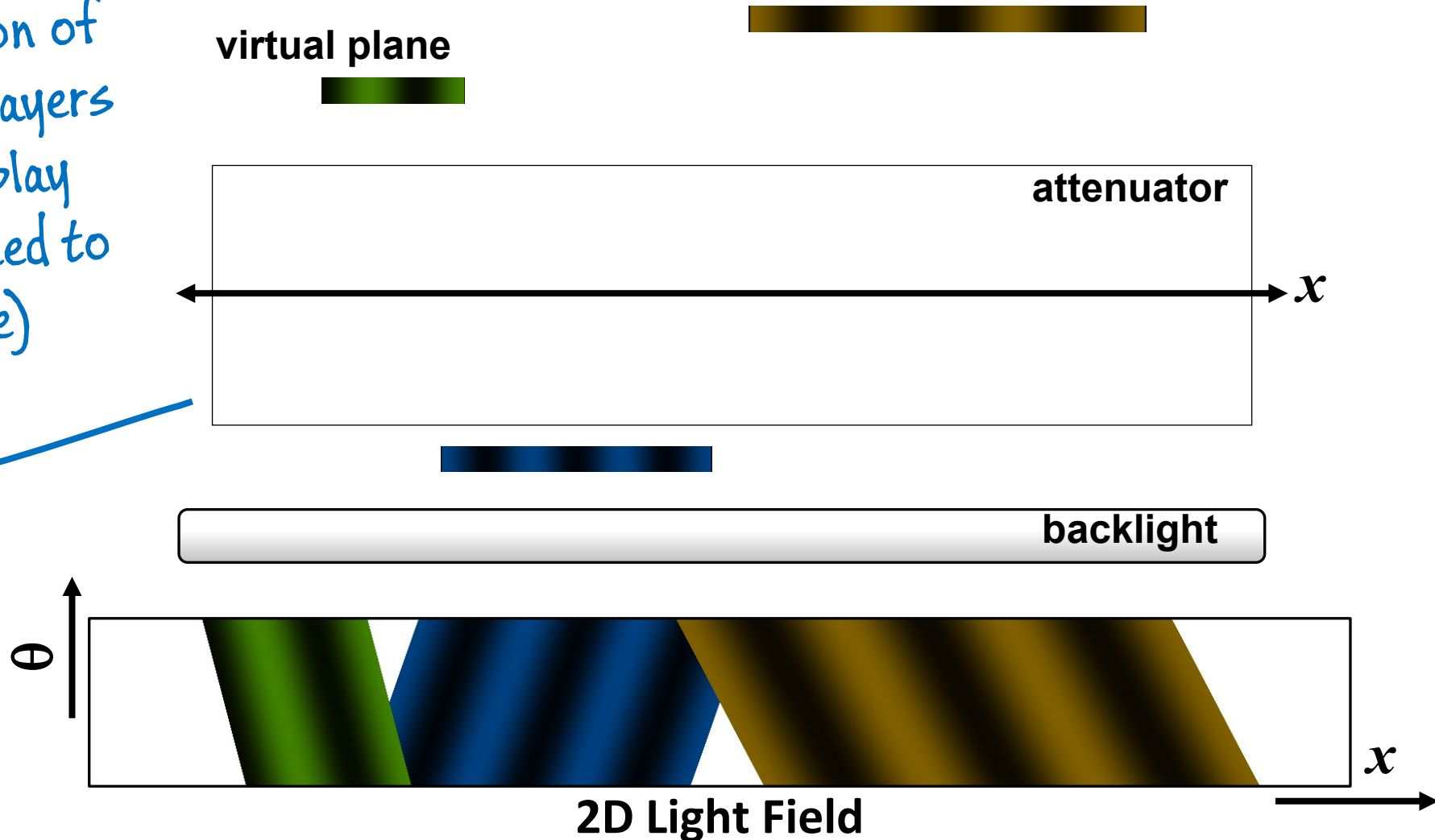
Light field visualization



...and assume this is the light field we want to show on the display.

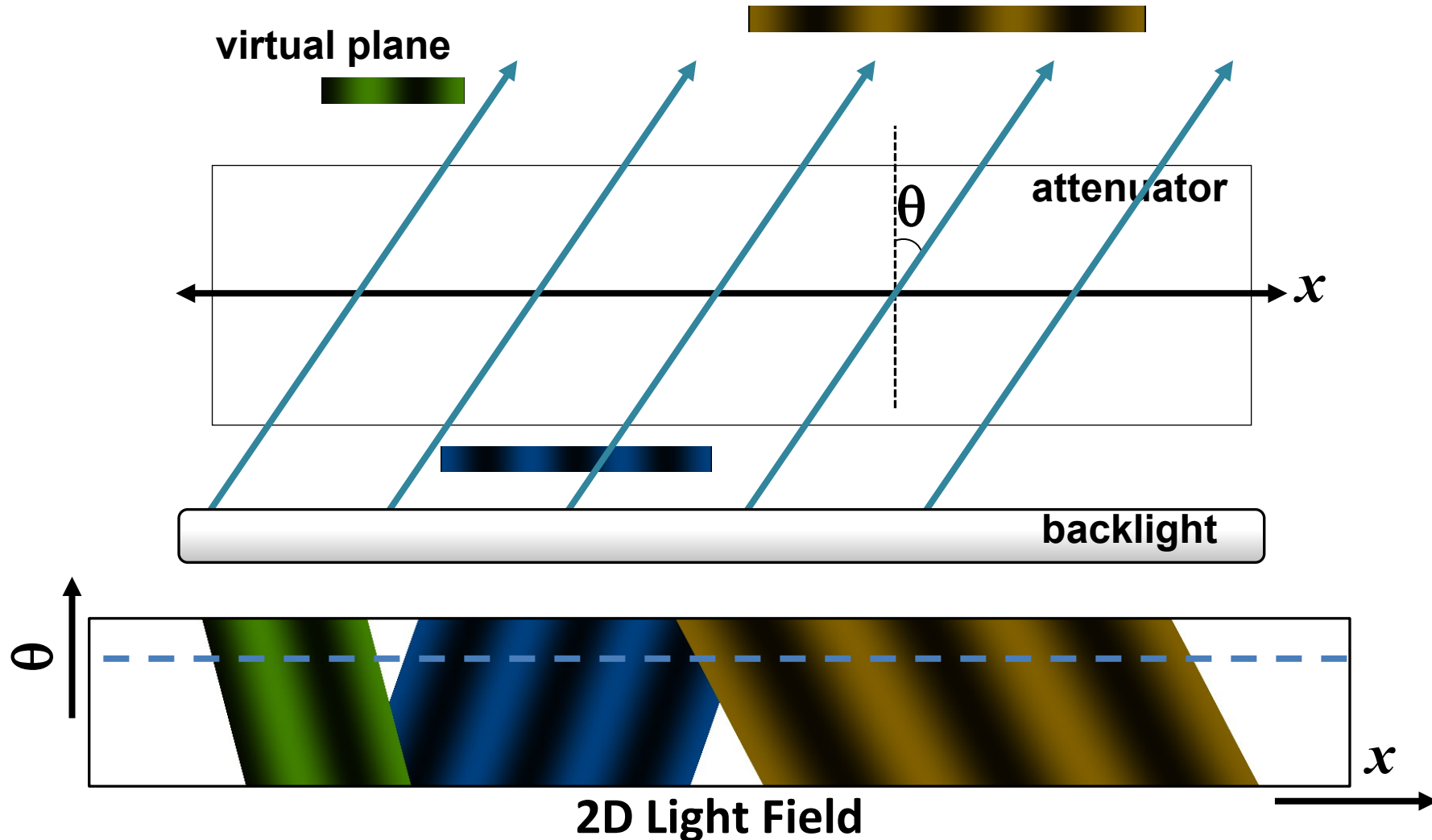
Tomographic light field synthesis

Cross-section of
the multiple layers
of the display
(what we need to
compute)



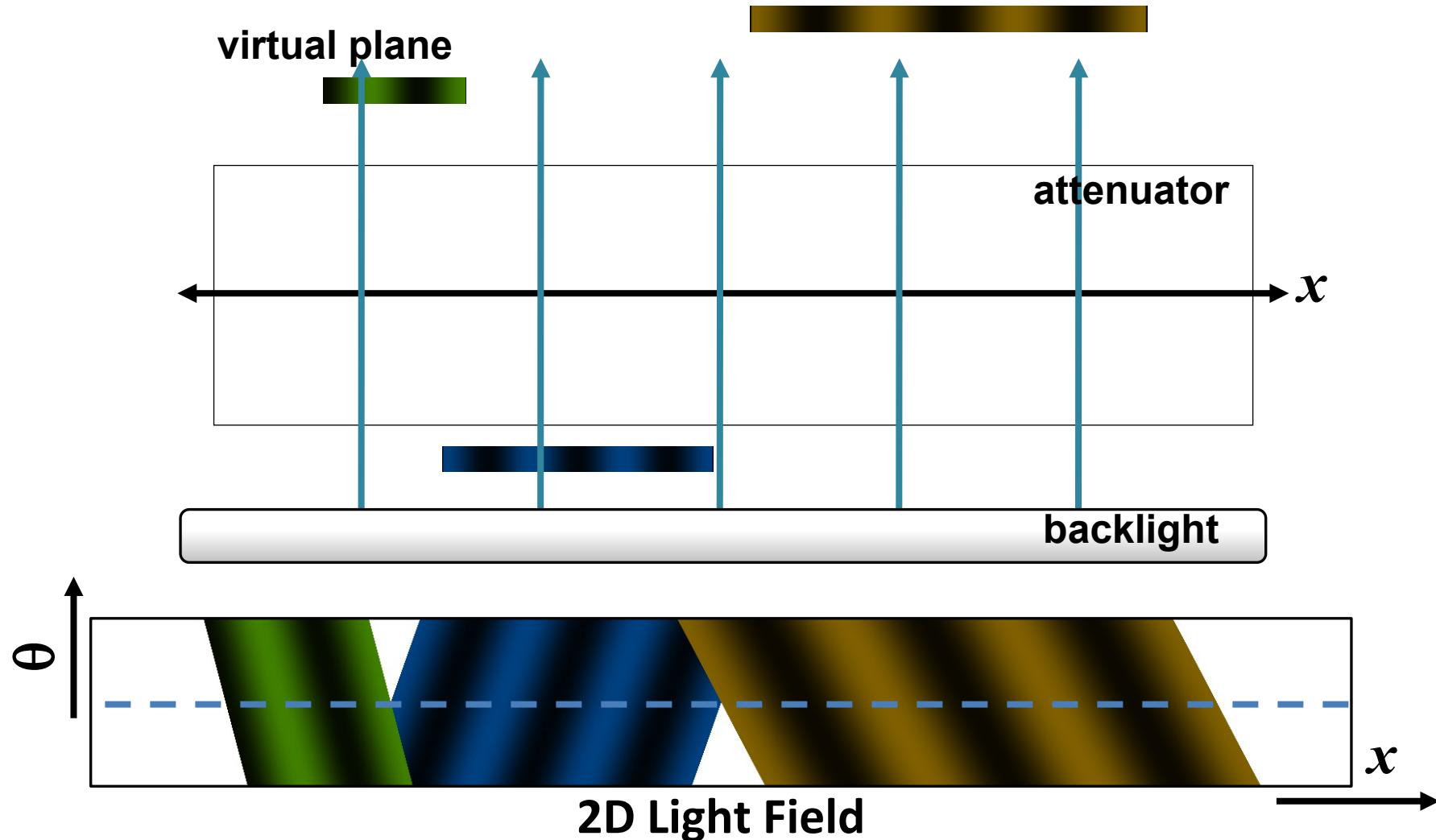
[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Tomographic light field synthesis



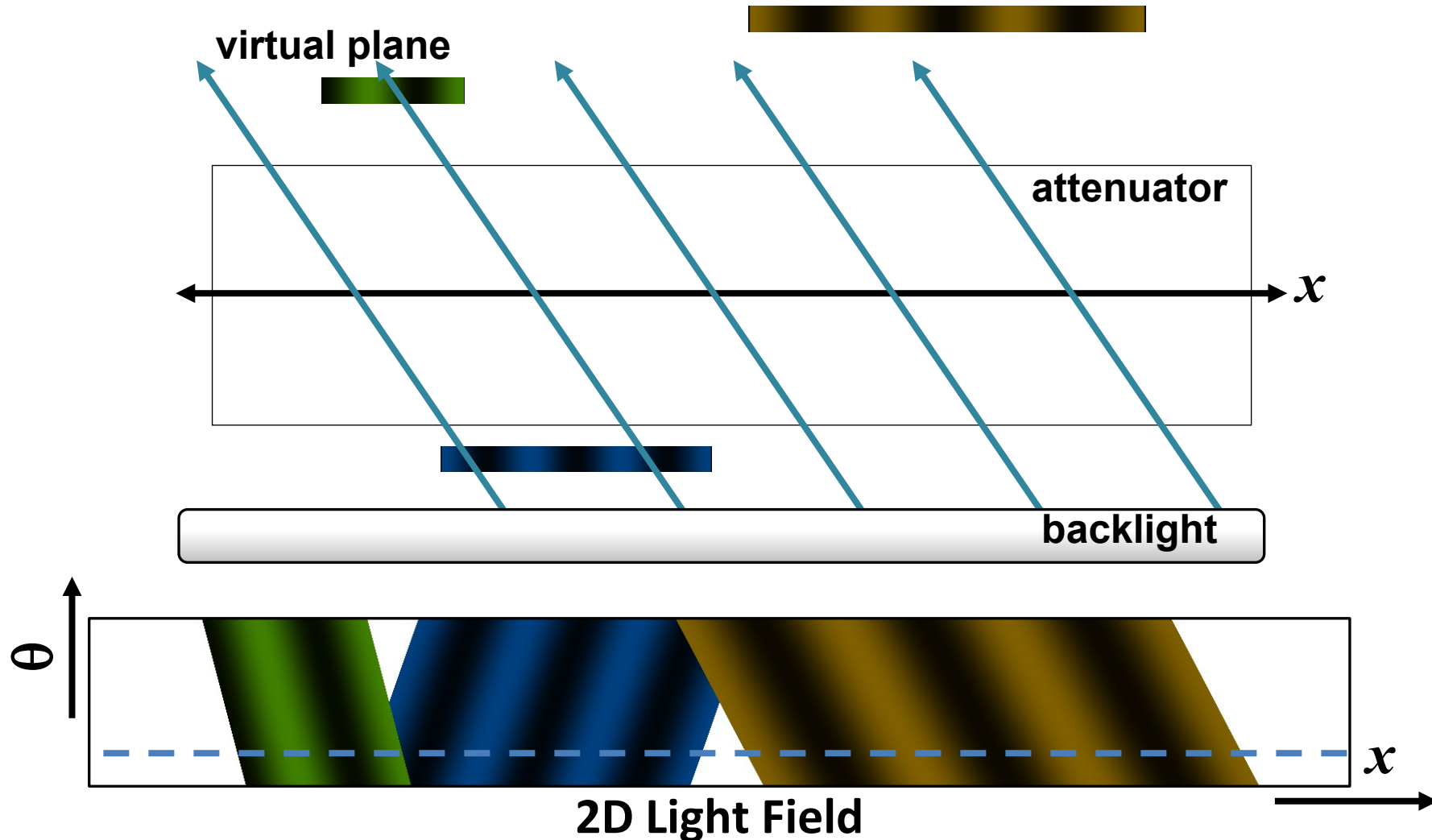
[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Tomographic light field synthesis



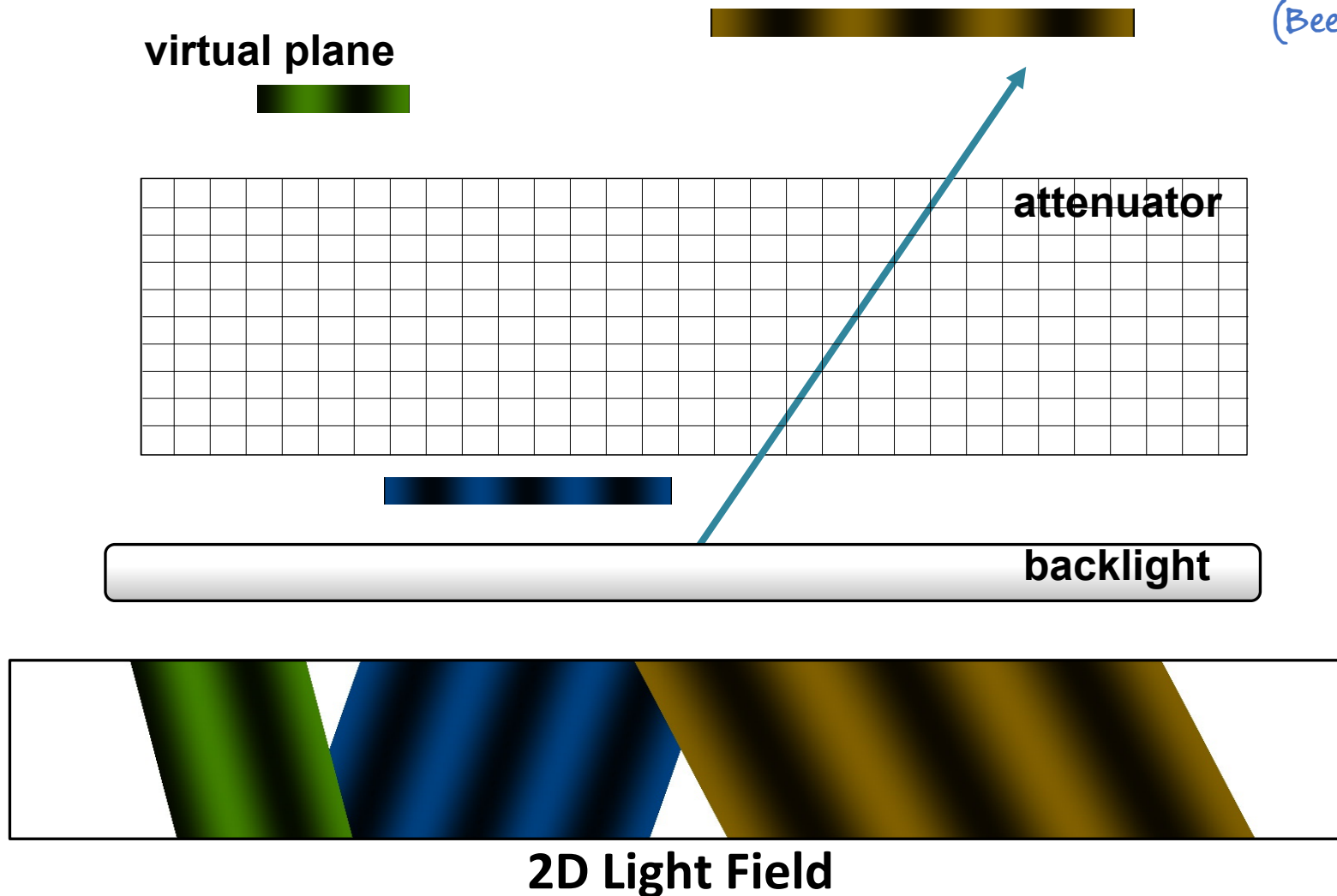
[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Tomographic light field synthesis



[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Tomographic light field synthesis



Intensity of transmitted light ray (Beer-Lambert law)

Incident intensity

Attenuation

Image formation model:

$$L(x, \theta) = I_0 e^{-\int_c \mu(r) dr}$$

$$\bar{L}(x, \theta) = \ln \left(\frac{L(x, \theta)}{I_0} \right) = - \int_c \mu(r) dr$$

Transmitted light field

$$\bar{\mathbf{I}} = -\mathbf{Pa}$$

Tomographic synthesis:

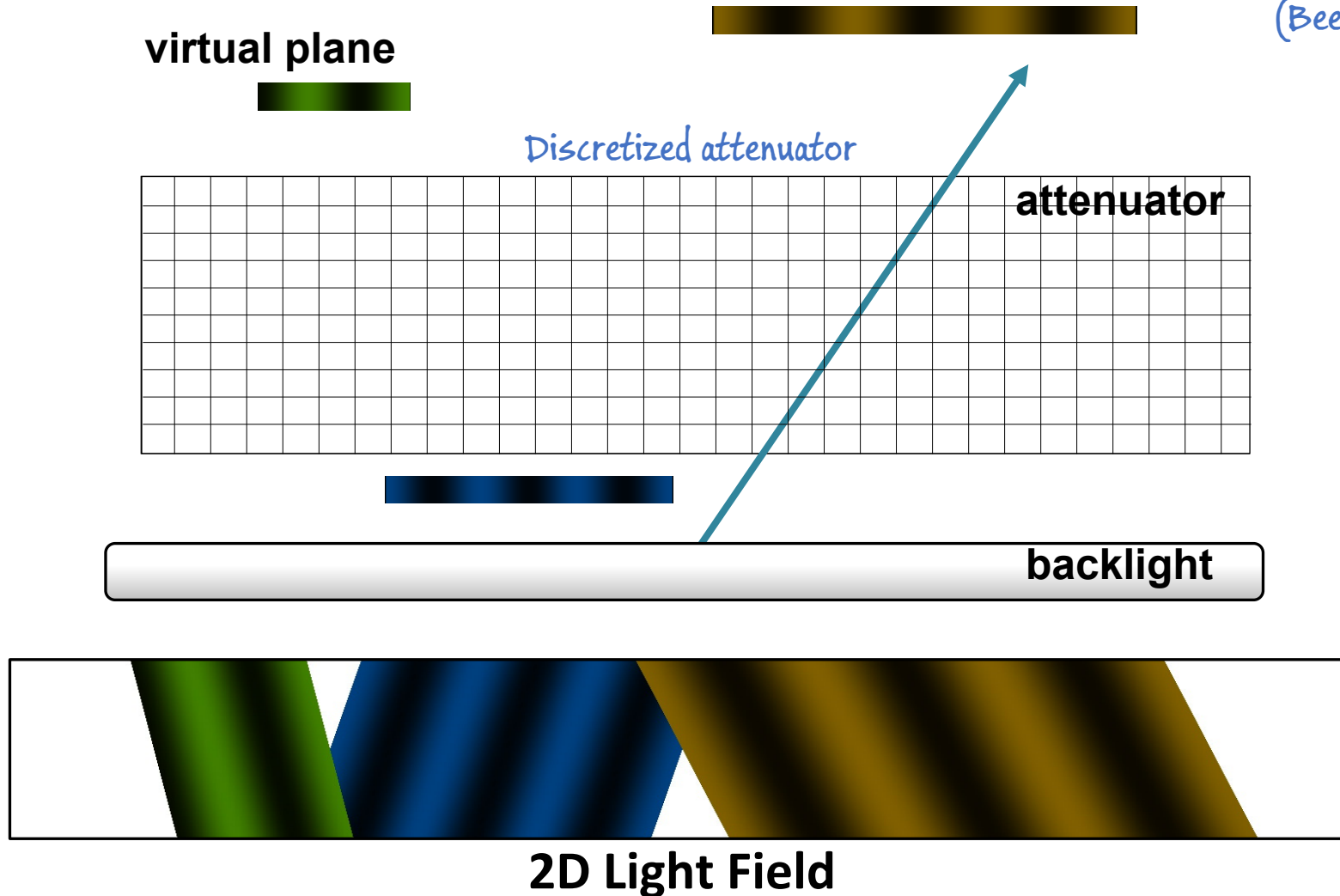
$$\arg \min_{\mathbf{a}} \left\| \bar{\mathbf{I}} + \mathbf{Pa} \right\|^2, \text{ for } \mathbf{a} \geq 0$$

Projection matrix (see paper for deriv.)

Attenuation coeff.

[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Tomographic light field synthesis



Intensity of transmitted light ray (Beer-Lambert law)

Incident intensity

Attenuation

Image formation model:

$$L(x, \theta) = I_0 e^{-\int_c \mu(r) dr}$$

$$\bar{L}(x, \theta) = \ln \left(\frac{L(x, \theta)}{I_0} \right) = - \int_c \mu(r) dr$$

Transmitted light field

$$\bar{\mathbf{I}} = -\mathbf{Pa}$$

Tomographic synthesis:

$$\arg \min_{\mathbf{a}} \left\| \bar{\mathbf{I}} + \mathbf{Pa} \right\|^2, \text{ for } \mathbf{a} \geq 0$$

Projection matrix (see paper for deriv.)

Attenuation coeff.

[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Tomographic light field synthesis

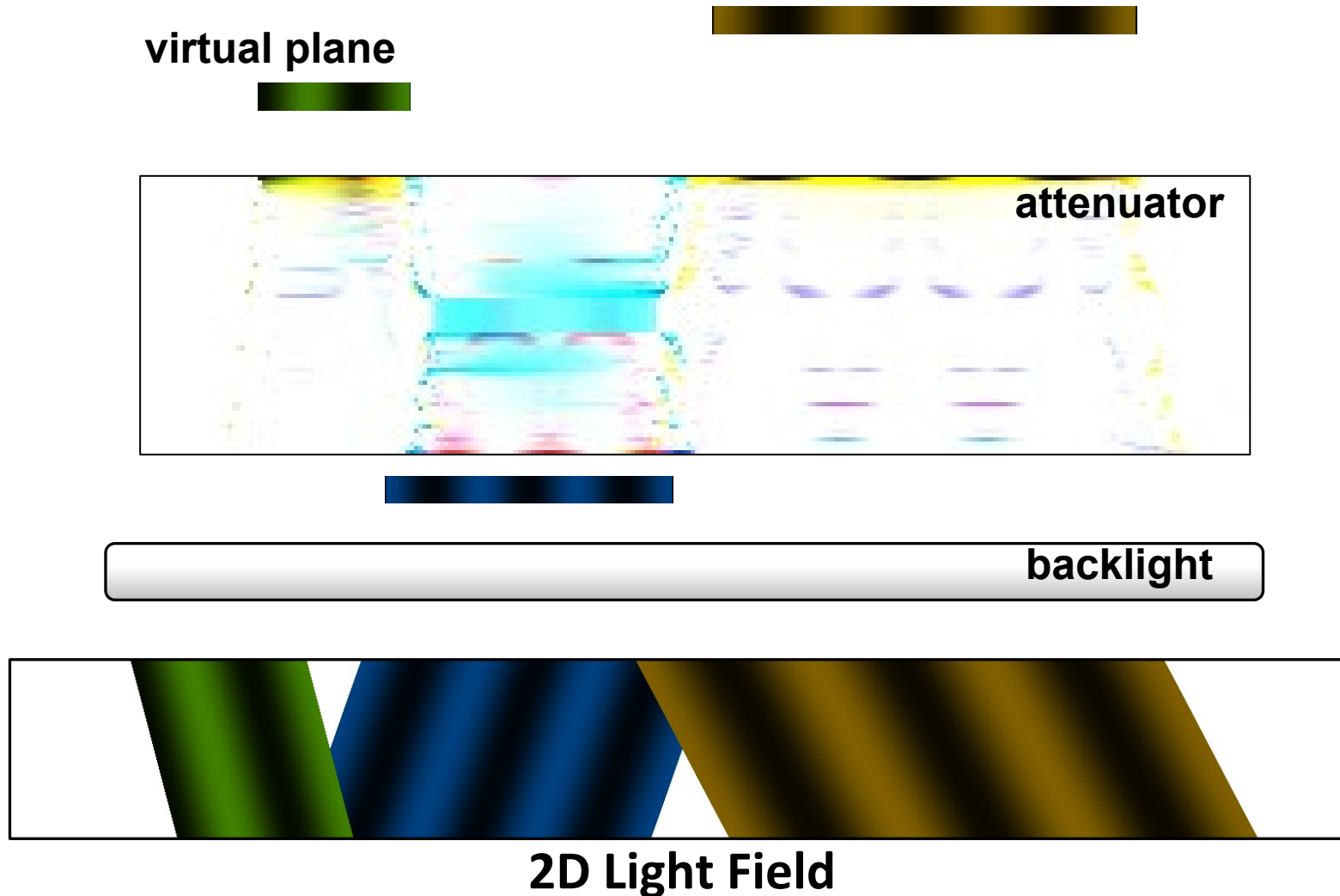


Image formation model:

$$L(x, \theta) = I_0 e^{-\int_c \mu(r) dr}$$

$$\bar{L}(x, \theta) = \ln \left(\frac{L(x, \theta)}{I_0} \right) = - \int_c \mu(r) dr$$

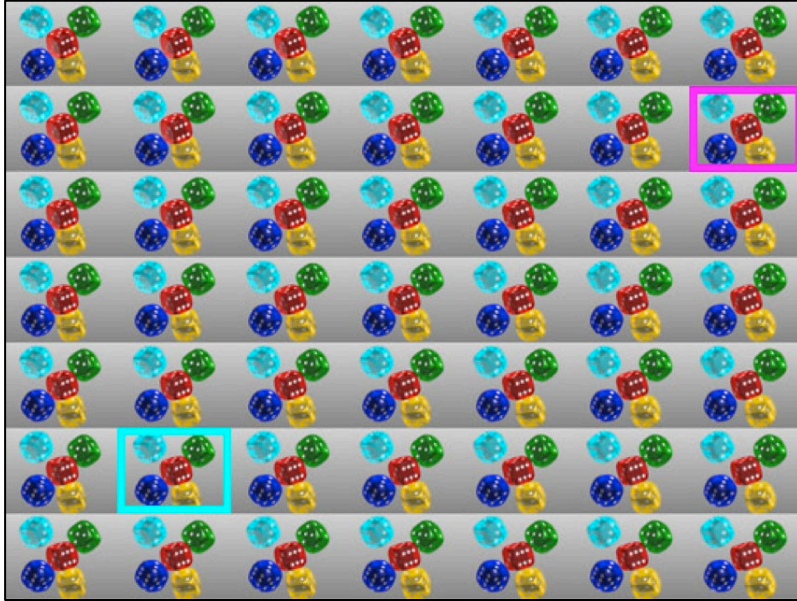
$$\bar{\mathbf{I}} = -\mathbf{P}\mathbf{a}$$

Tomographic synthesis:

$$\arg \min_{\mathbf{a}} \left\| \bar{\mathbf{I}} + \mathbf{P}\mathbf{a} \right\|^2, \text{ for } \mathbf{a} \geq 0$$

[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

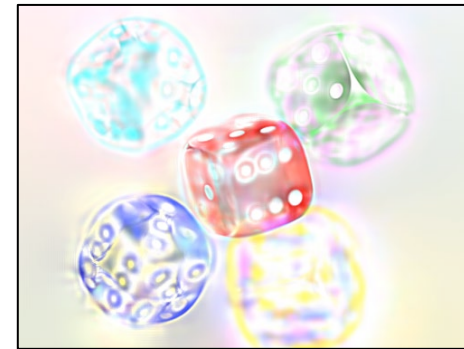
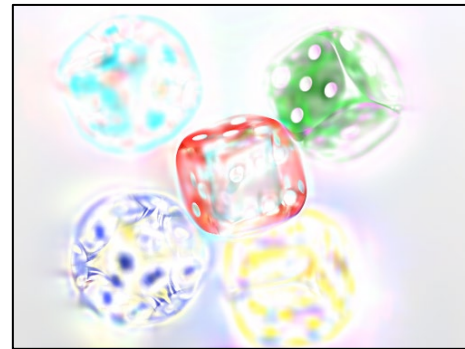
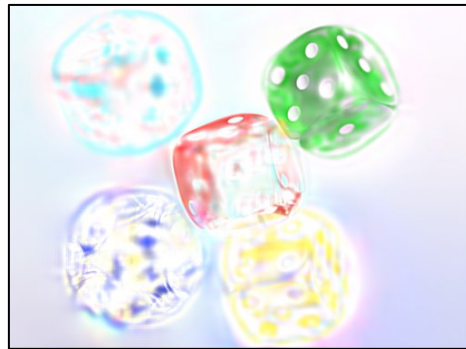
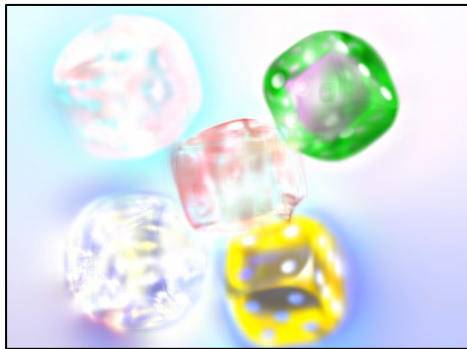
Multi-layer Light Field Decomposition



Target 4D Light Field



Reconstructed Views



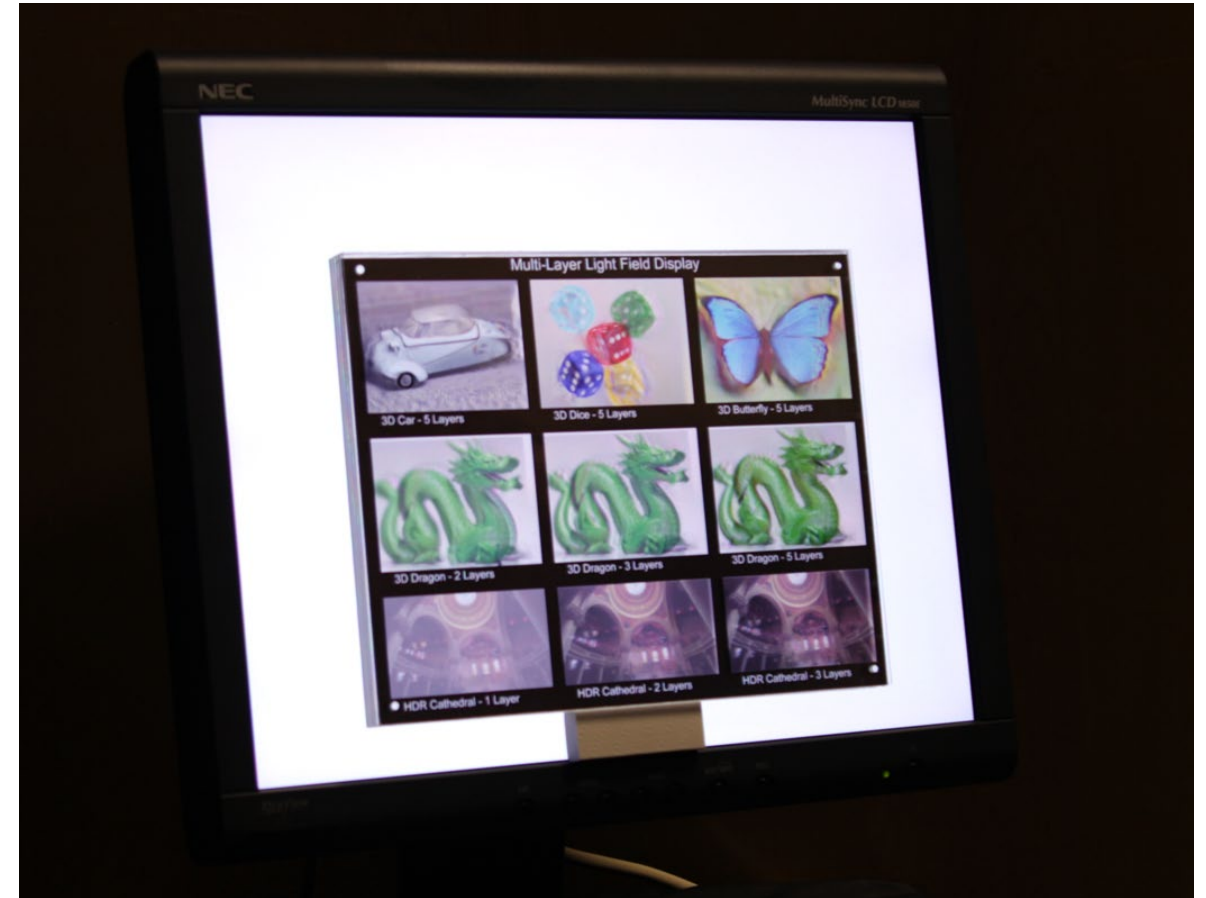
Multi-Layer Decomposition

[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

Prototype multi-layer display: Layered 3D



Transparency stack with acrylic spacers



Prototype in front of LCD (backlight source)

[Wetzstein et al. 2011] Layered 3D. SIGGRAPH.

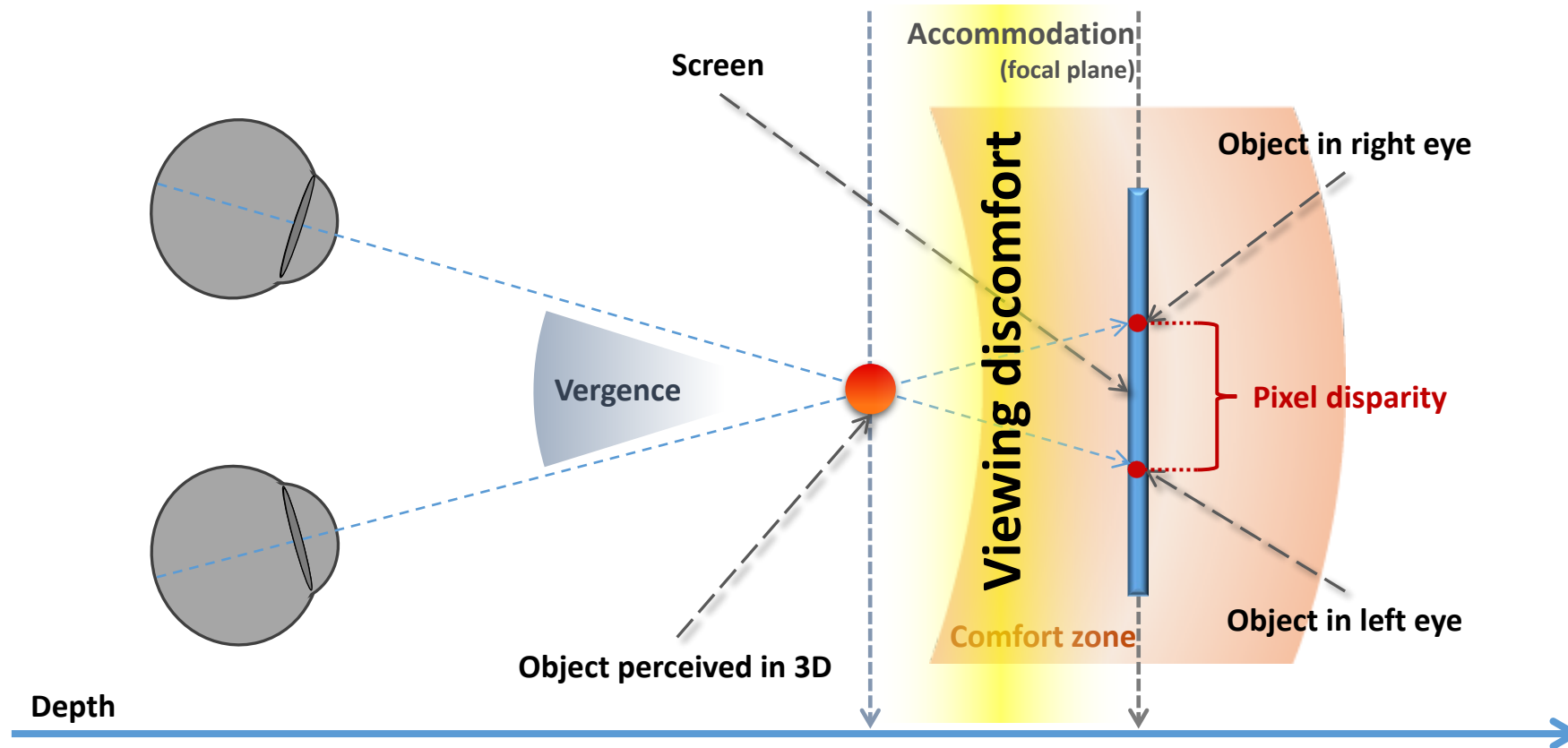


Stereo and light field displays – Limitations

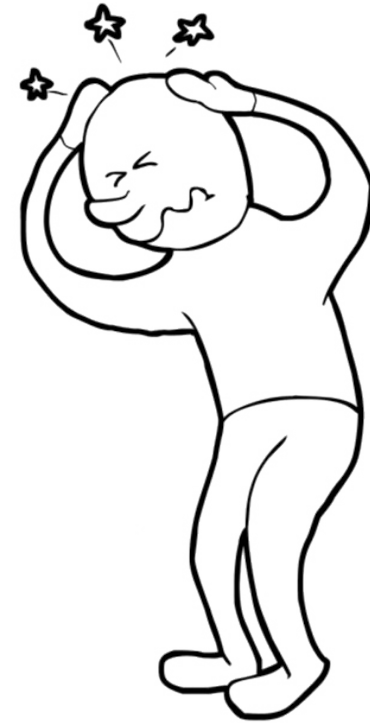
- Brightness
- Field of view
- Spatial resolution
- Angular resolution
- Form factor
- Moiré, crosstalk
- Vergence-accommodation conflict
- Depth of field

Critical in most current 3D displays

Vergence-accommodation conflict (VAC) – Why?



Consequence of VAC: Viewing discomfort



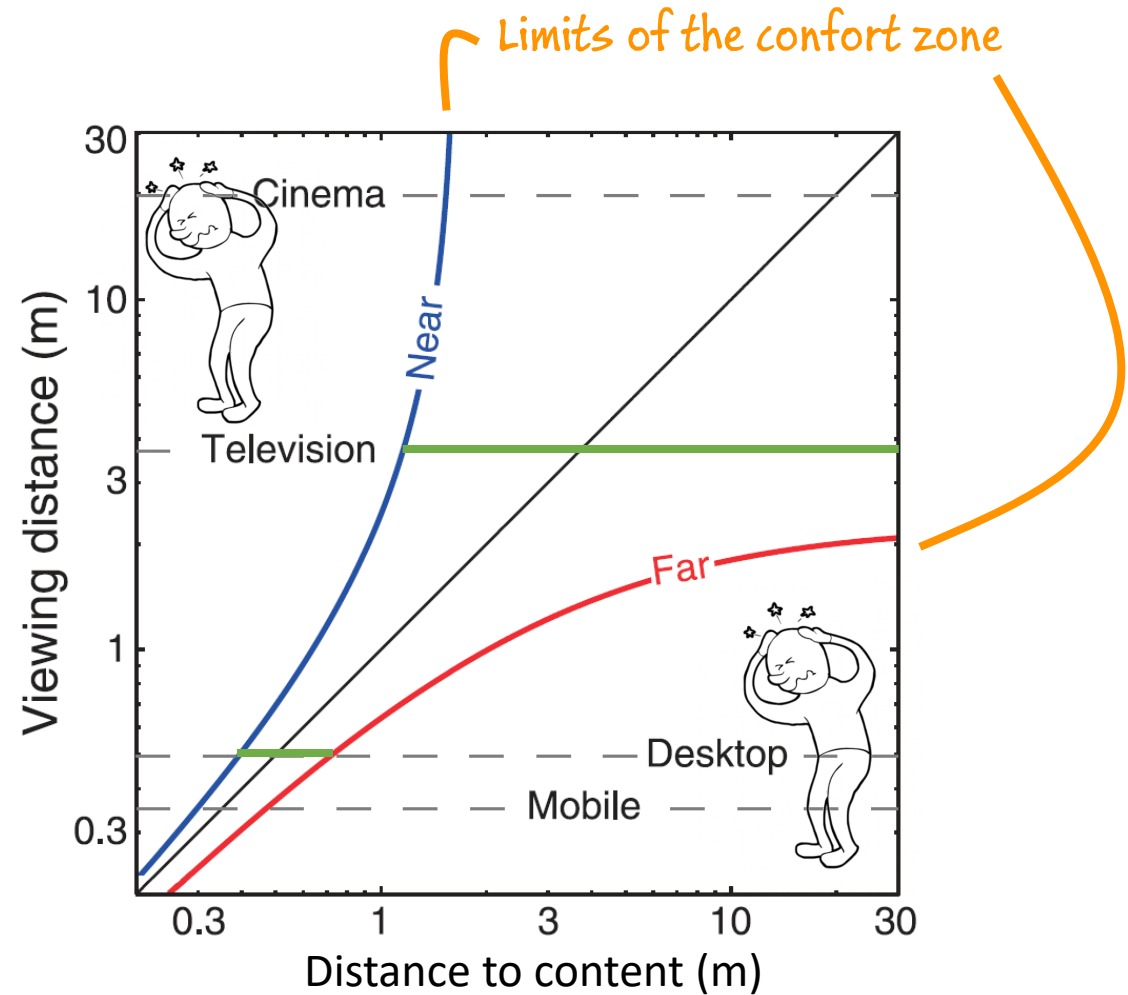
VAC – Comfort zones have been measured

Comfort zone size depends on:

- Presented content
- Viewing condition
- Screen distance

Other factors:

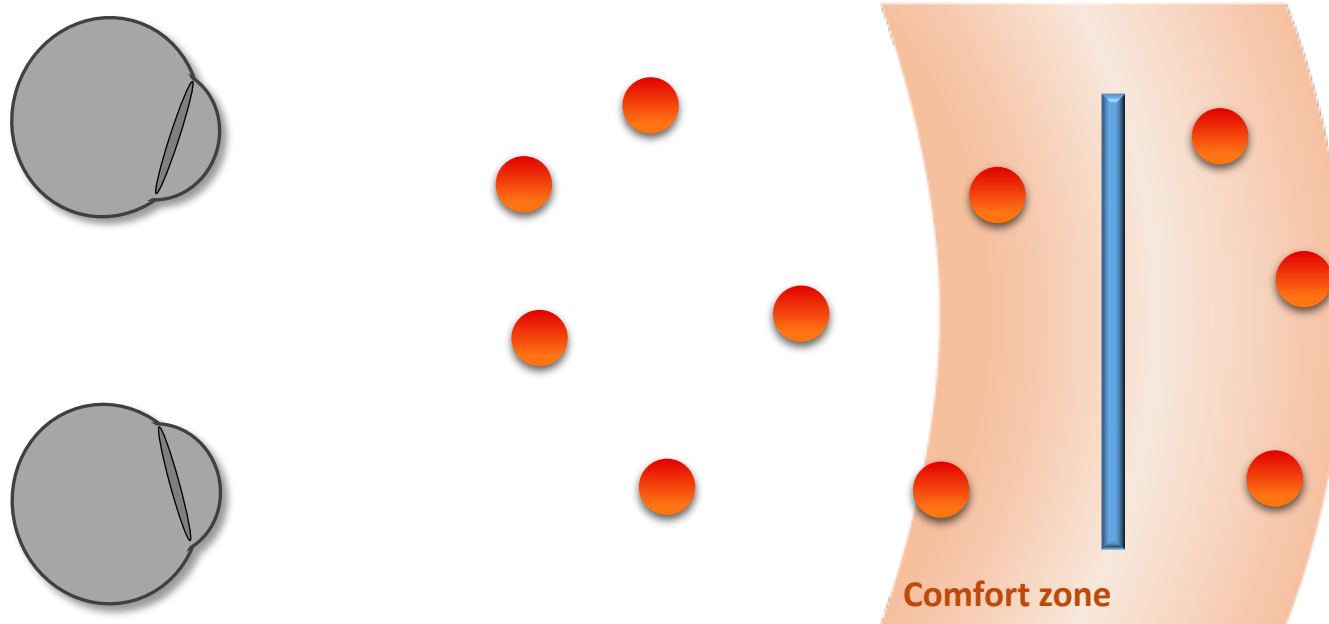
- Distance between eyes
- Depth of field
- Temporal coherence



"The zone of comfort: Predicting visual discomfort with stereo displays" by Shibata et al. 2011

Source: P. Didyk. Stereo Retargeting. Multidimensional Image Retargeting Course. SIGGRAPH Asia 2011.

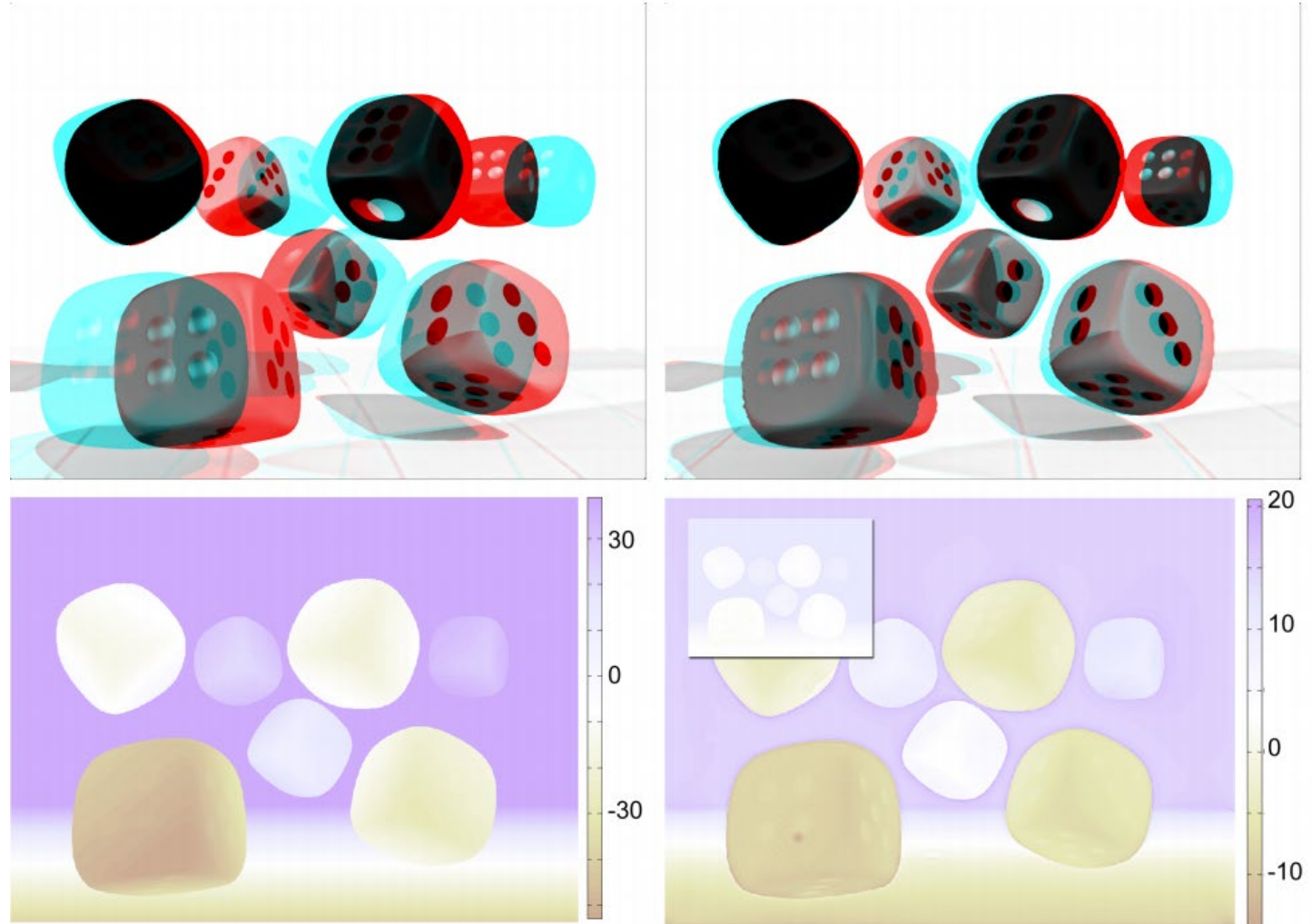
A solution to VAC: Depth manipulation



Scene manipulation
Viewing discomfort → **Viewing comfort**

A solution to VAC: Depth manipulation

- There are a number of techniques that manipulate depth so that it lies within the comfort zone.



A solution to VAC: Depth manipulation

- There are a number of techniques that manipulate depth so that it lies within the comfort zone.

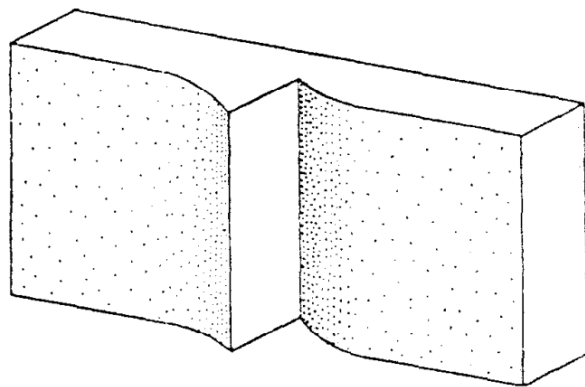
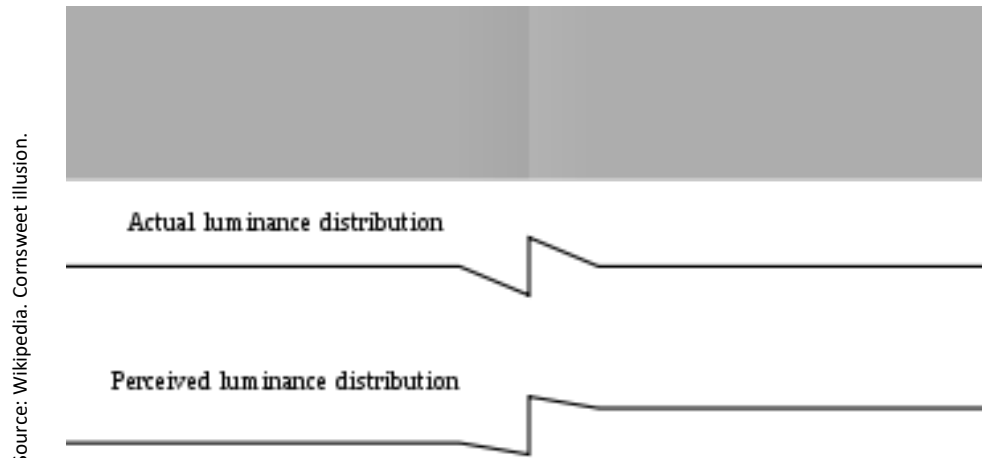
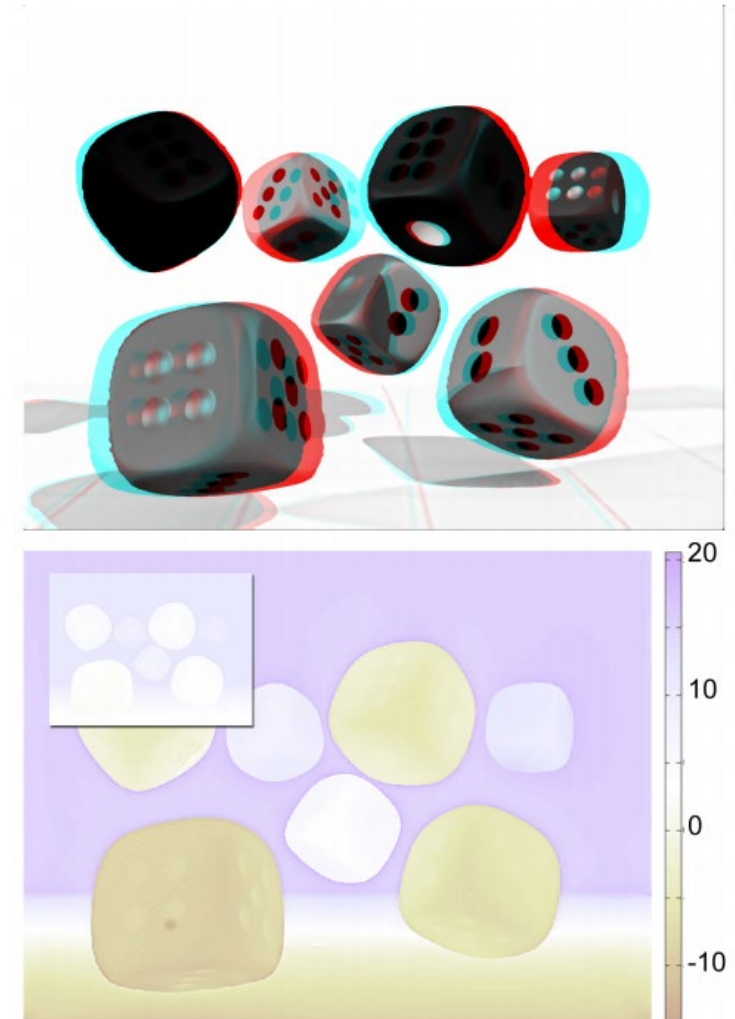


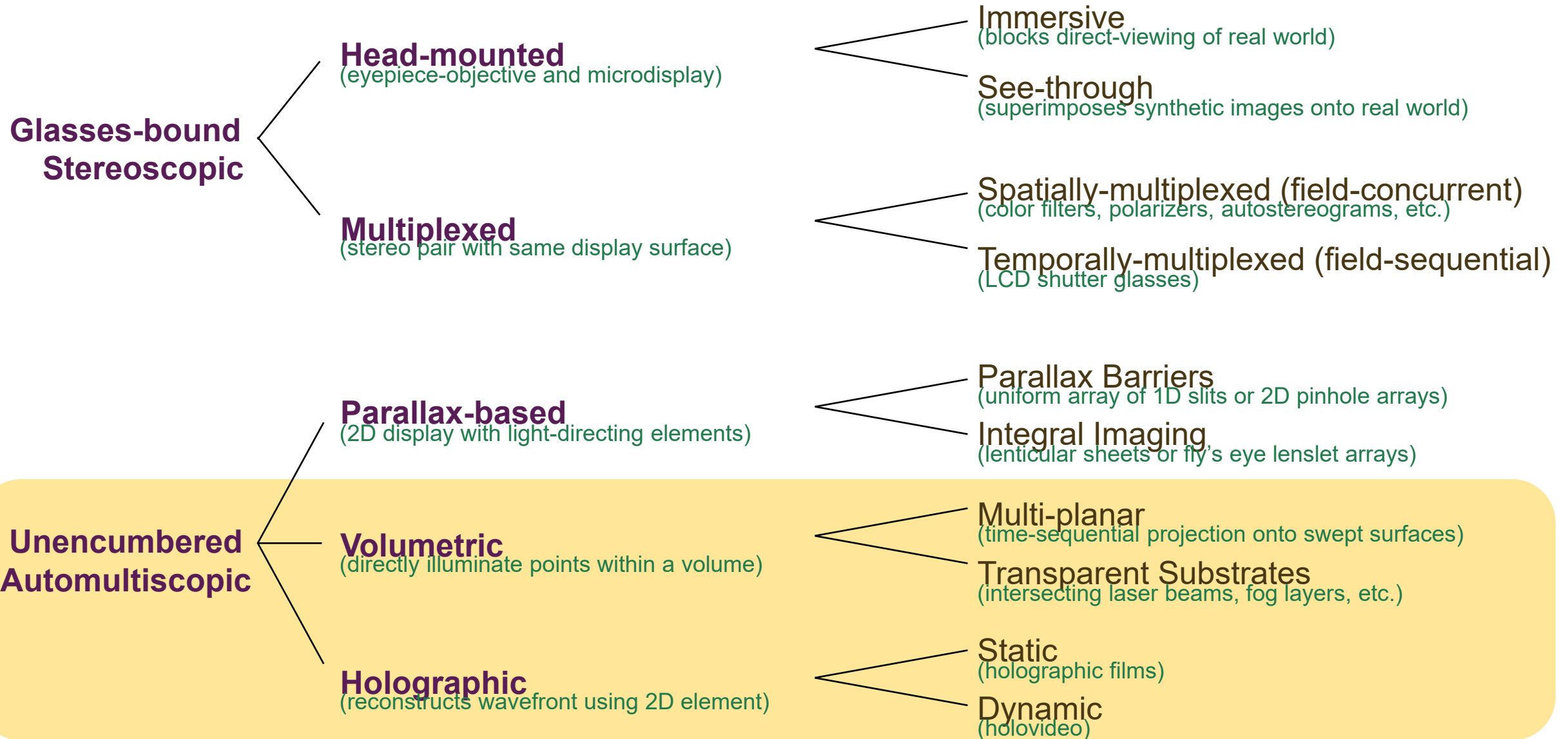
Fig. 4. Perspective sketch of the illusory depth surface. Left part looks apparently nearer than the right part.

Craik-O'Brien-Cornsweet illusion for luminance

Craik-O'Brien-Cornsweet illusion for visual depth



A Taxonomy of 3D Displays



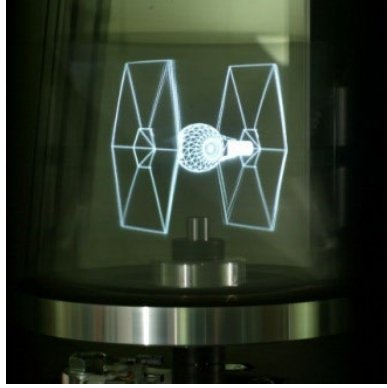
Volumetric & Holographic Displays



LightSpace



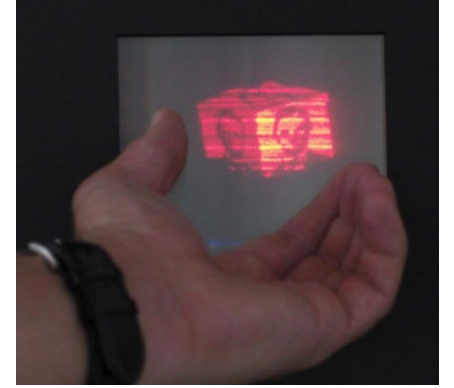
Sony



Jones et al. 2007



Zebra Imaging



MIT Hologvideo

Volumetric Displays

Holograms

- 3D objects only inside enclosure
- mostly mechanically moving parts
- all depth cues
- computationally expensive

Take-home messages

Capture and display share underlying principles.

There is often cross-pollination between them.

The combination of optics, computation (and perception) can yield improvements at different stages of the imaging pipeline.

Capture, processing or editing, display (and even manufacture!)

Knowledge of human perception can help overcome inherent limitations of hardware.

Joint co-design of hardware, and software, taking also into account perception.

What else is out there?

Computational Ophthalmology – Tailored Displays



Strong Presbyopia
(farsightedness)



Tailored
Glasses-Free
Display



(a) Input Image

(b) Regular Display

(c) Tailored Display

Computational Ophthalmology



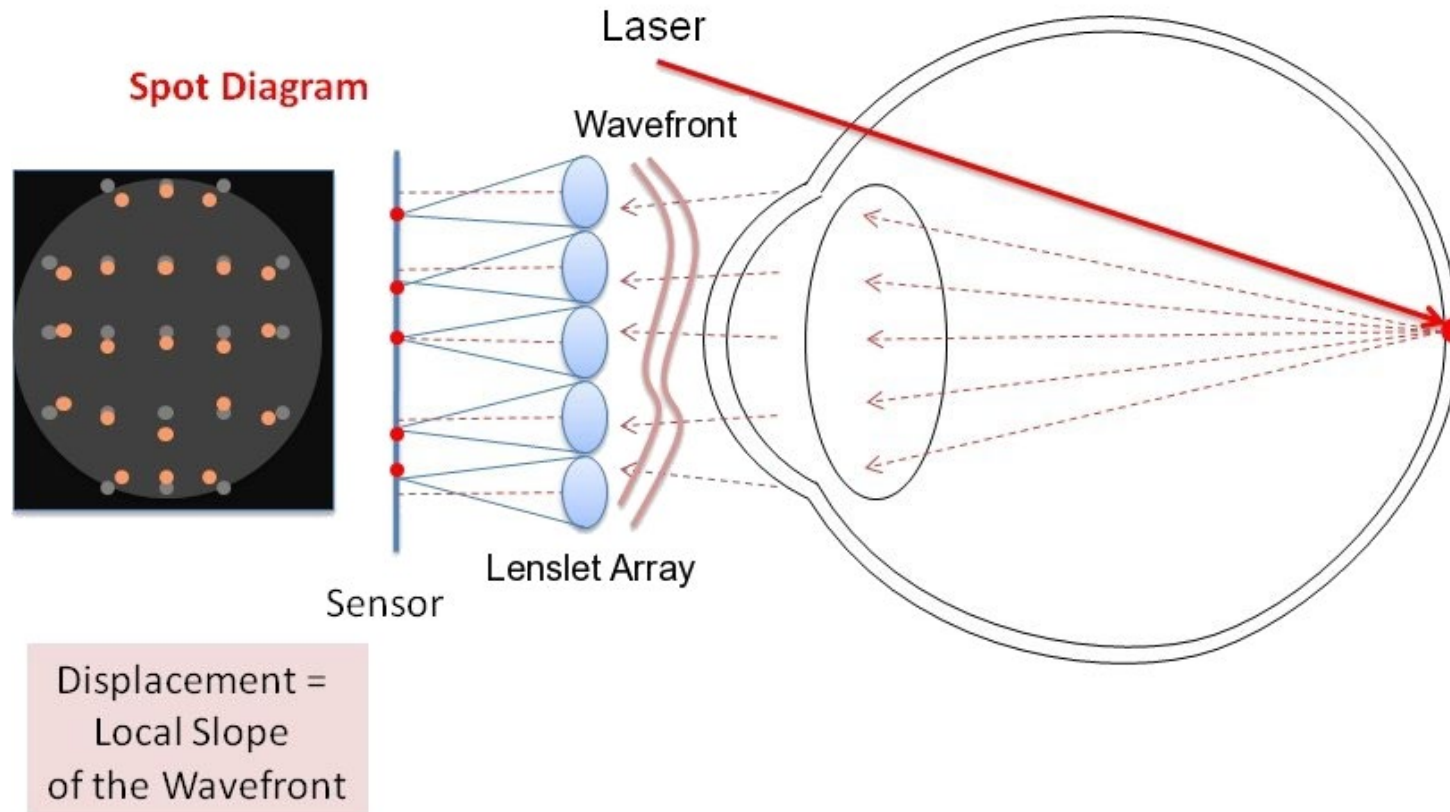
[Pamplona 2010]



Computational Ophthalmology – Refractive Errors



Inspired on Shack-Hartmann wavefront sensors, used to characterize an imaging system. Used, among others, in clinical optics.

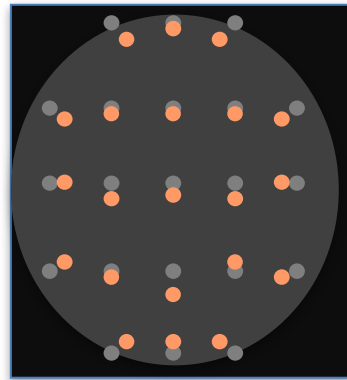


Computational Ophthalmology – Refractive Errors

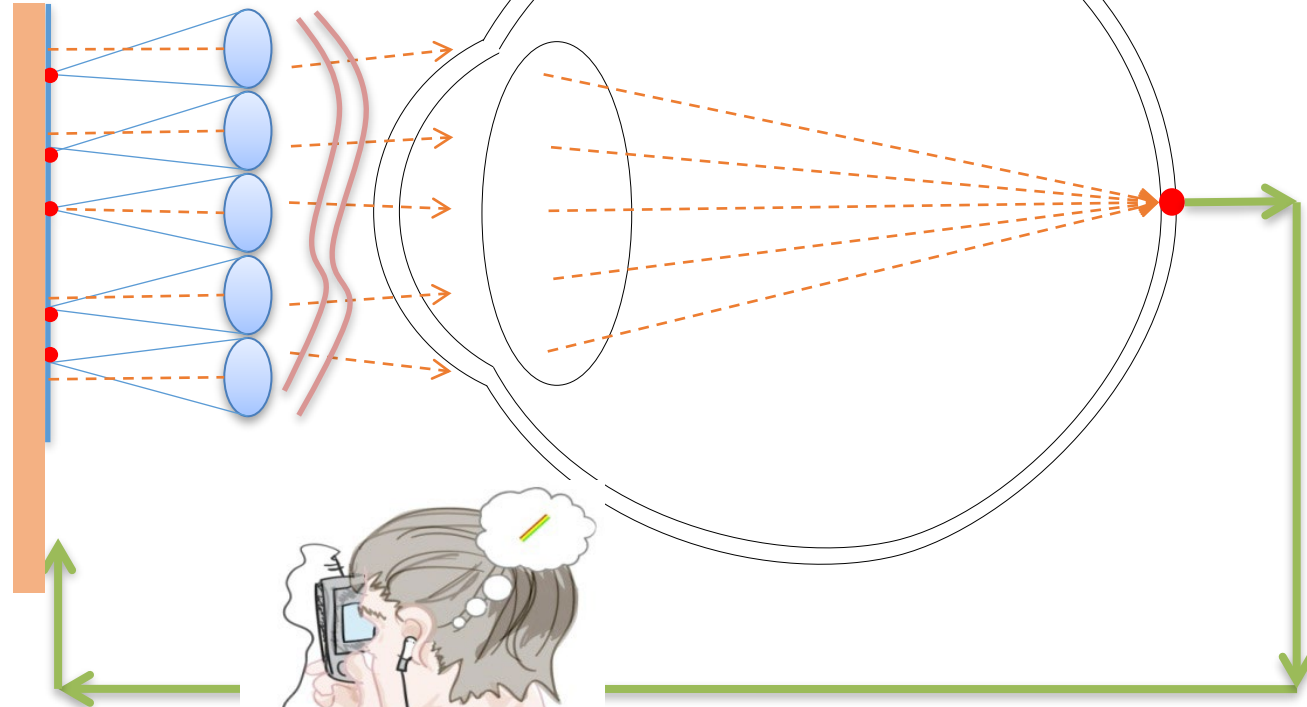


Inverse of Shack-Hartmann, *user interactive!*

Spot Diagram
on LCD

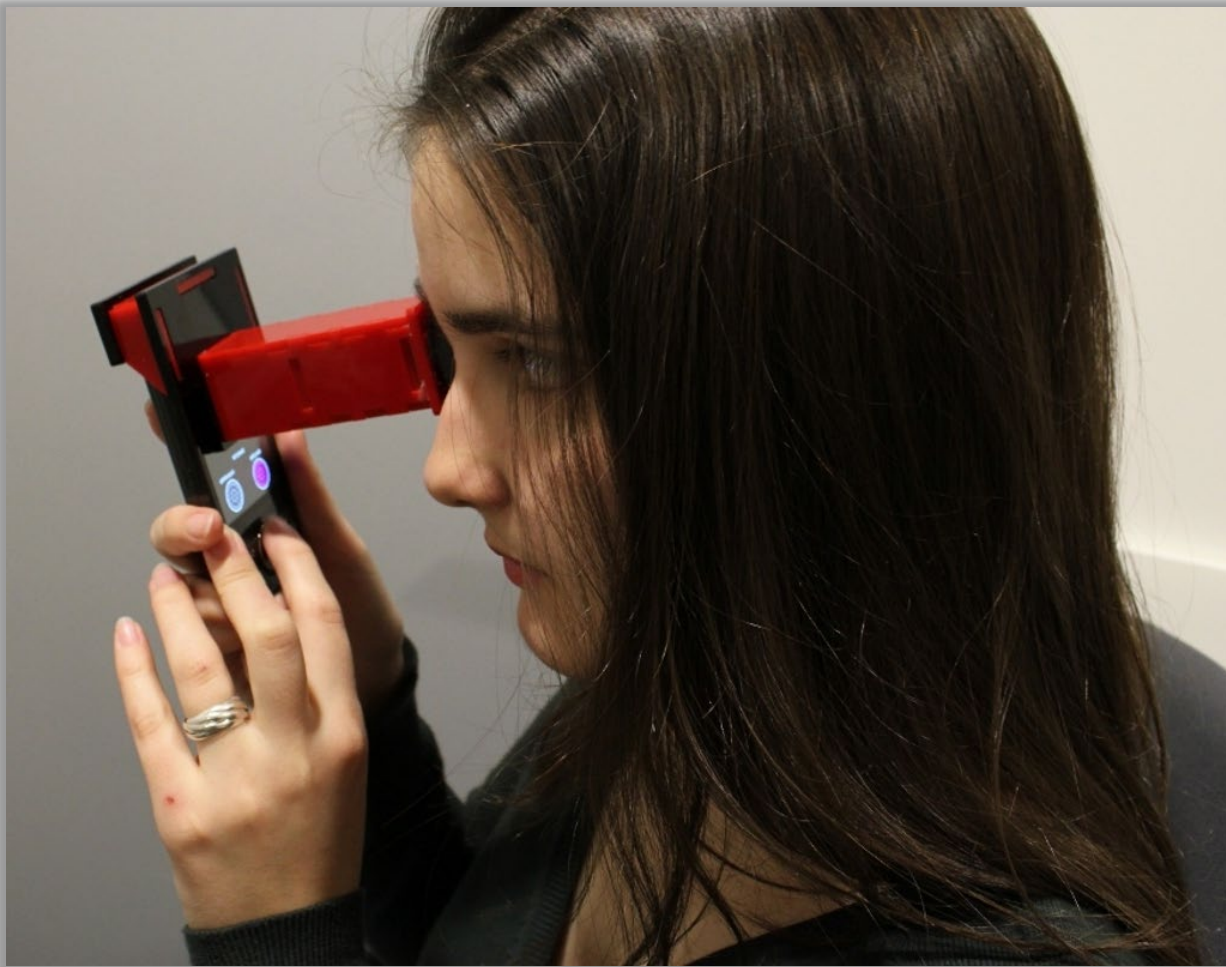


CellPhone
LCD



Displace 25
points but
3 parameters

Computational Ophthalmology - Cataracts



Computational Ophthalmology - Cataracts

